# Coilem A/B/C/D benchmark

## PASS for supported cases; deferred hardware coverage (v0.5.0)

Harness 1.7.0; protocol 1.4.0.

Report presentation: 1.7.0.

Candidate commit: 5b7cad88c43c0ac7ade9265988f71b6f2ba2884f

Reference: FROZEN_REFERENCE; commit 738d9516e8500c5996c772bd96470472cbb9a0f9.

Orange solid: Coilem; blue dashed: saved FEMM. No reference solves executed.

| Phase | Recorded outcomes |
|---|---|
| A – SPM | 10 PASS |
| B – IPM | 12 PASS |
| C – Distributed Winding | 1 Deferred hardware coverage (v0.5.0), 4 PASS |
| D – BLDC | 3 PASS |

Phase summary: green for all active PASS; yellow for mixed or missing active coverage; red for more than 50% active FAIL. Deferred hardware rows are grey. Raw statuses, counts and the overall numerical status retain the complete selection. Legacy ORNL Prius coverage is distinct from the Pyleecan software benchmark.

## Phase A – SPM results

| Fixture | Outcome |
|---|---|
| spm_4p12s_simple__baseline | PASS |
| spm_4p12s_simple__high_current | PASS |
| spm_4p12s_simple__tight_airgap | PASS |
| spm_4p12s_simple_noload__baseline | PASS |
| spm_4p12s_simple_noload__high_current | PASS |
| spm_4p12s_simple_noload__tight_airgap | PASS |
| spm_8p12s_concentrated__baseline | PASS |
| spm_8p12s_concentrated__high_current | PASS |
| spm_8p12s_concentrated__tight_airgap | PASS |
| spm_8p12s_reference_angle30__baseline | PASS |

## Phase B – IPM results

| Fixture | Outcome |
|---|---|
| ipm_flat_12s10p_fscw_contract | PASS |
| ipm_flat_24s16p_industrial_contract | PASS |

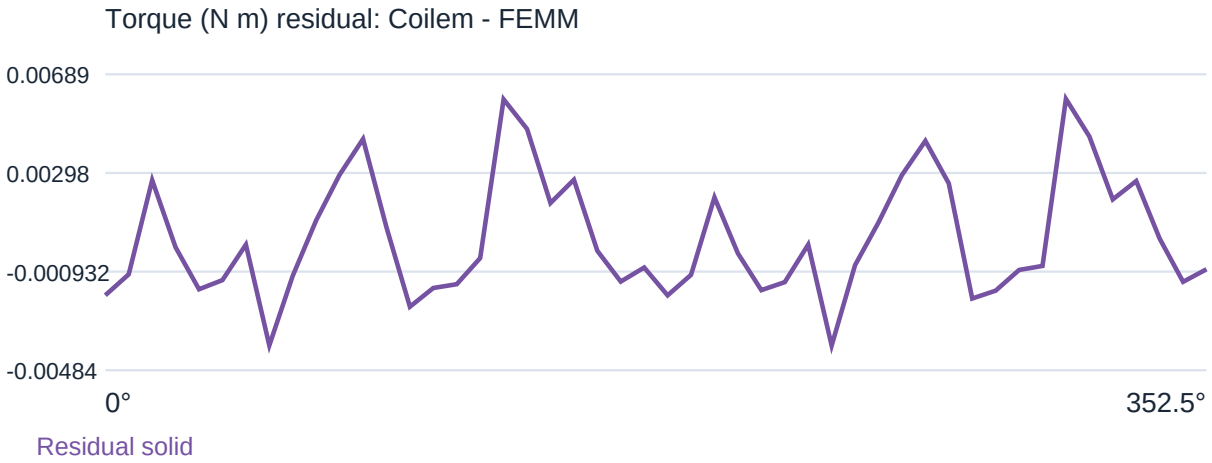
| Fixture | Outcome |
|---|---|
| ipm_flat_4p12s_micro_contract | PASS |
| ipm_flat_8p12s_fsae_angled_pocket_contract | PASS |
| ipm_flat_8p12s_fsae_contract | PASS |
| ipm_flat_8p12s_fsae_thin_bridge_contract | PASS |
| ipm_vshape_8p12s_baseline_contract | PASS |
| ipm_vshape_8p12s_deep_apex_high_output_v1 | PASS |
| ipm_vshape_8p12s_narrow_angle_contract | PASS |
| ipm_vshape_8p12s_pocket_offset_contract | PASS |
| ipm_vshape_8p12s_thin_bridge_contract | PASS |
| ipm_vshape_8p12s_wide_angle_high_output_v1 | PASS |

## Phase C – Distributed Winding results

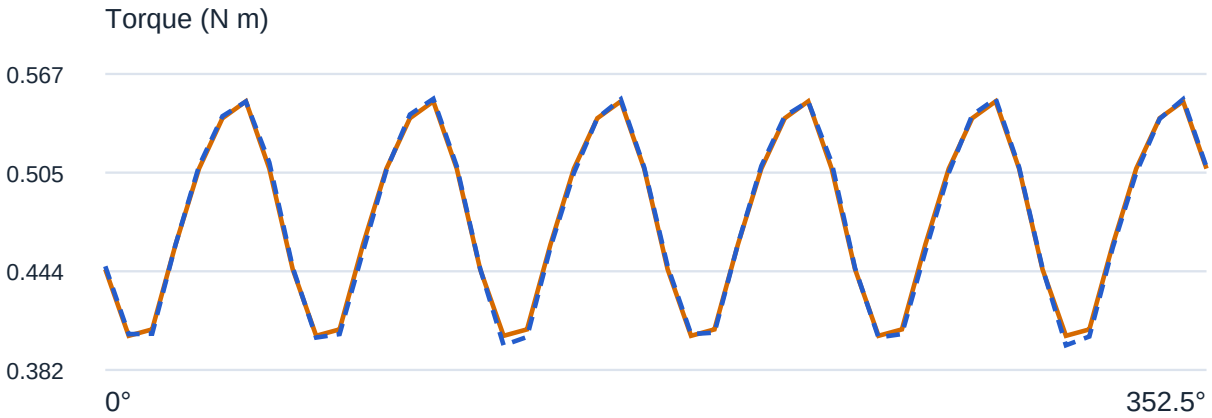
| Fixture | Outcome |
|---|---|
| ornl_prius_2004_ipm | Deferred hardware coverage (v0.5.0) |
| spm_12p36s_distributed_femm_validation | PASS |
| spm_8p24s_distributed_femm_validation | PASS |
| spm_8p48s_chorded_5_6 | PASS |
| spm_8p48s_distributed_femm_validation | PASS |

# Phase A – spm_4p12s_simple__baseline

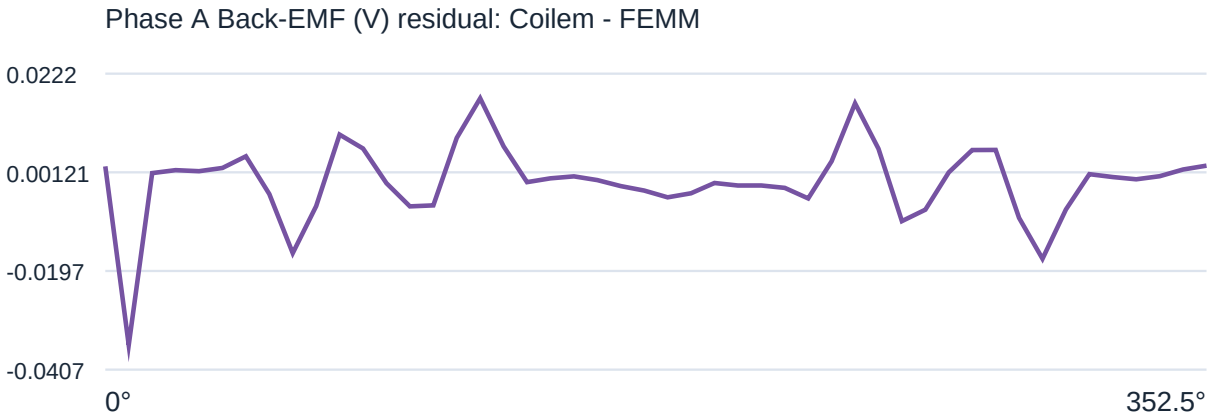
Frozen native fixture JSON



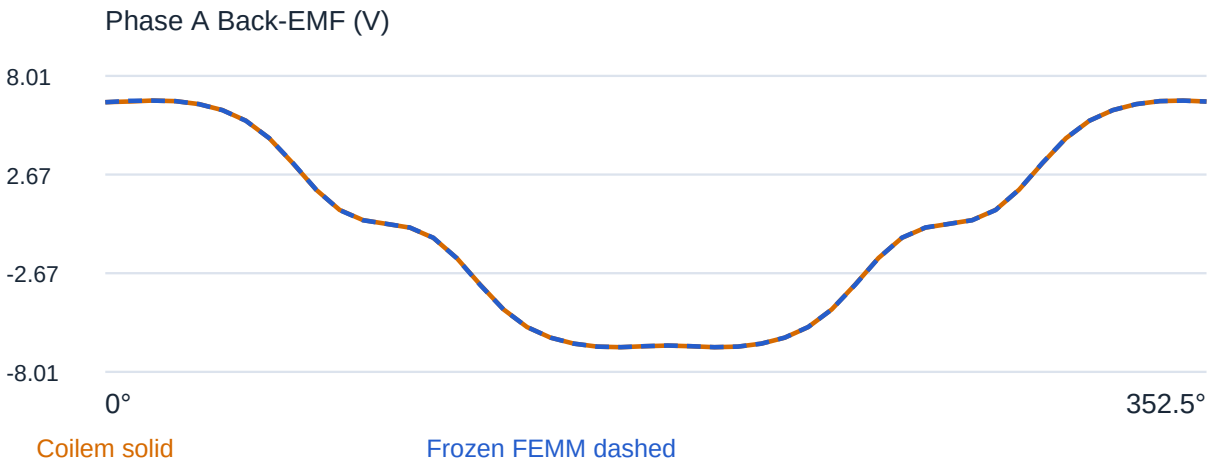

Residual solid




Coilem solid  Frozen FEMM dashed



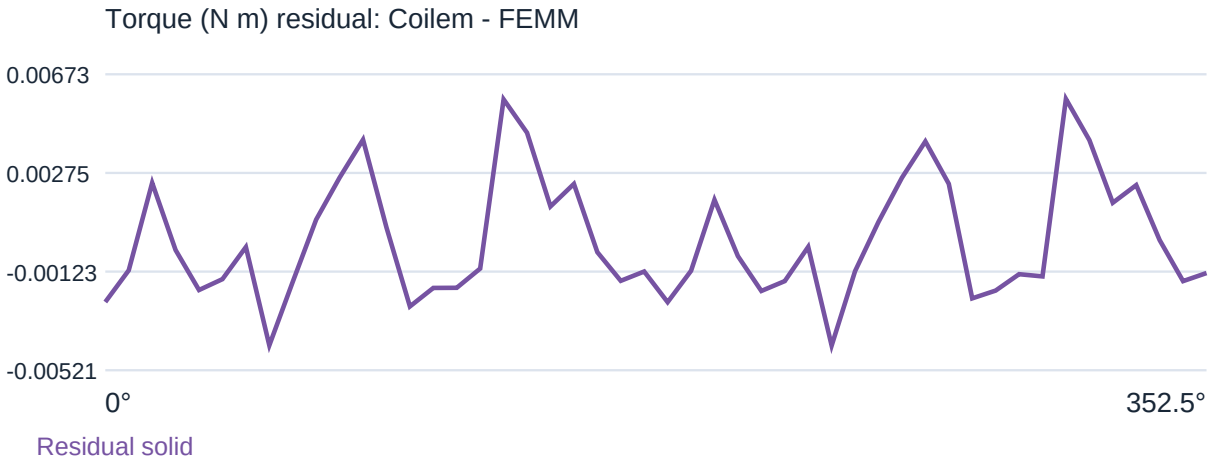

Residual solid

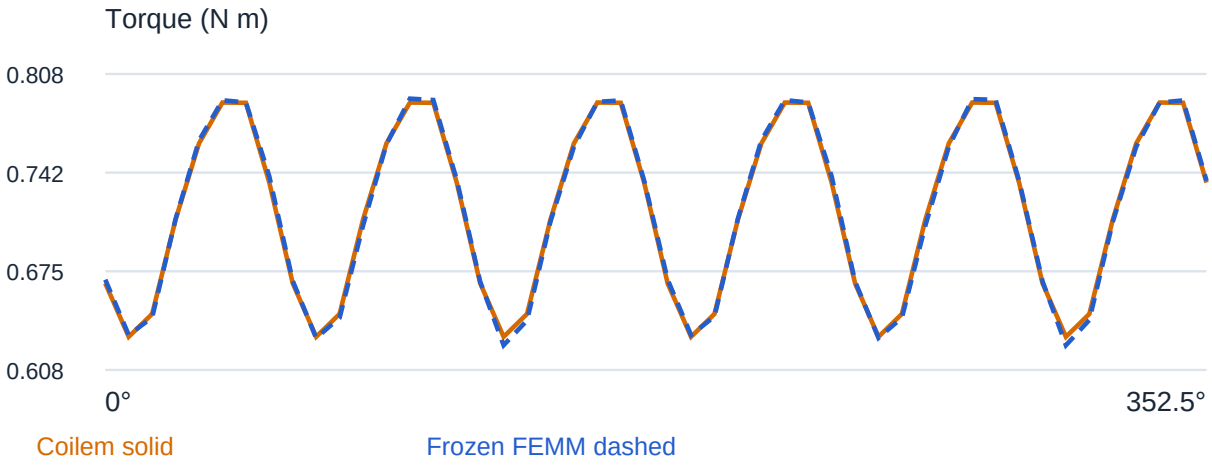


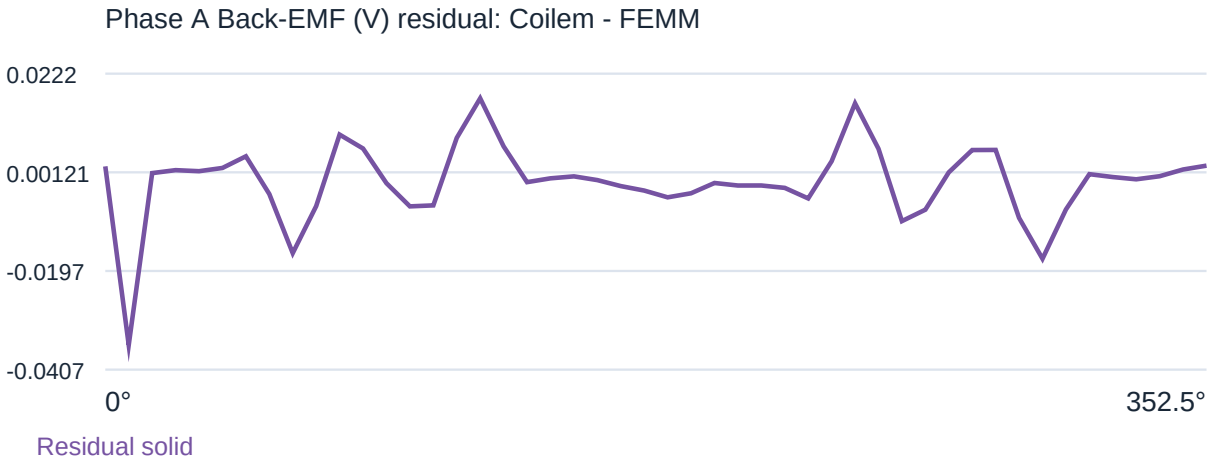

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

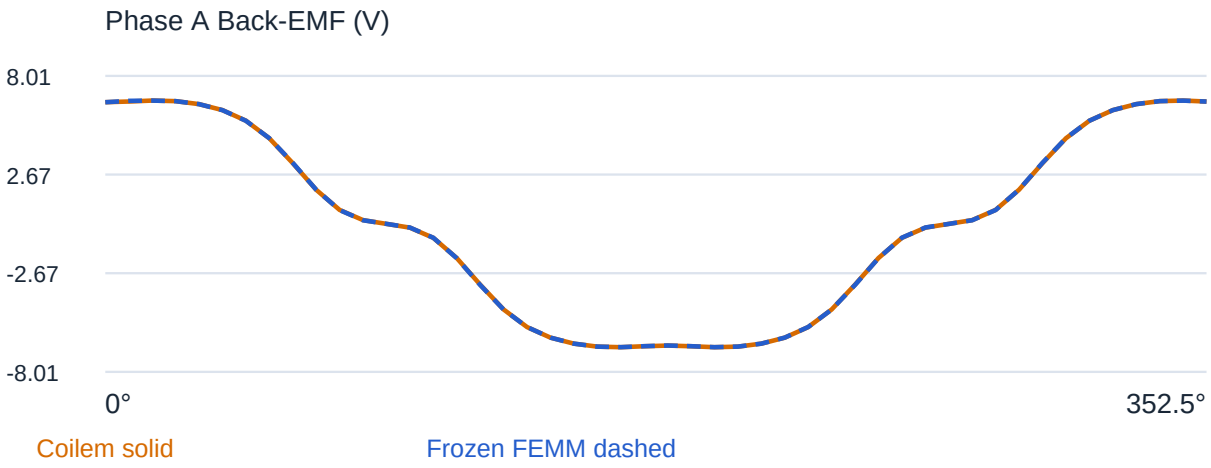
# Phase A – spm_4p12s_simple__high_current

Frozen native fixture JSON



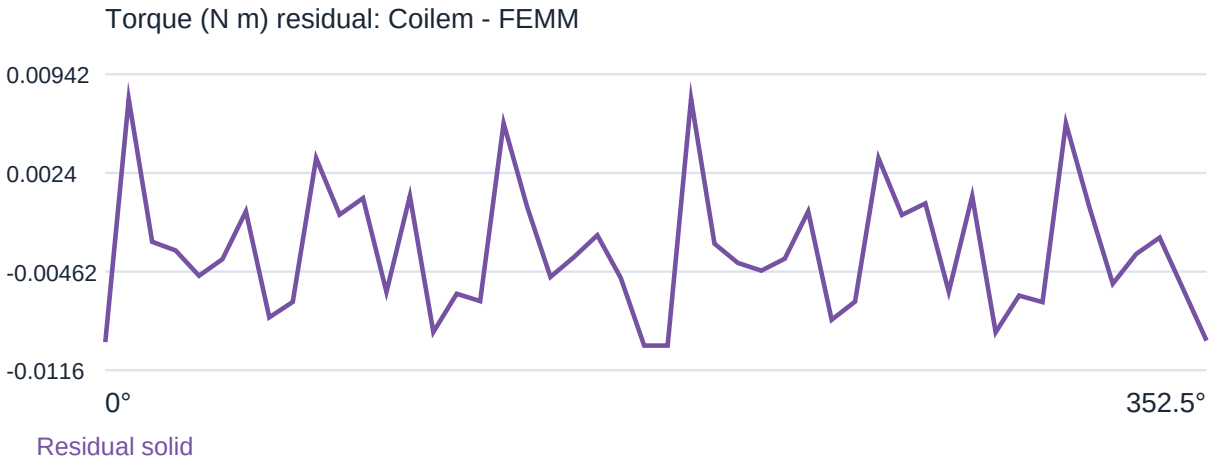





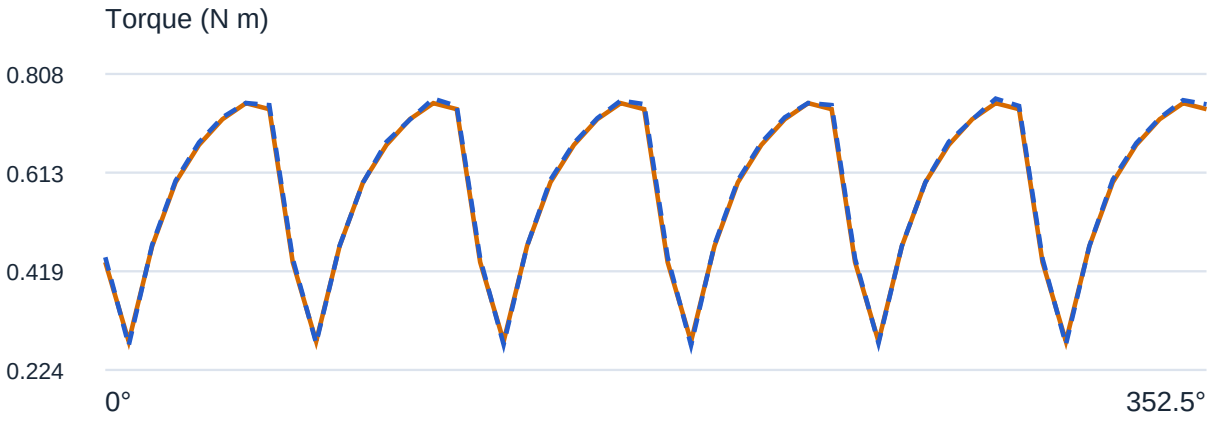

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase A – spm_4p12s_simple__tight_airgap

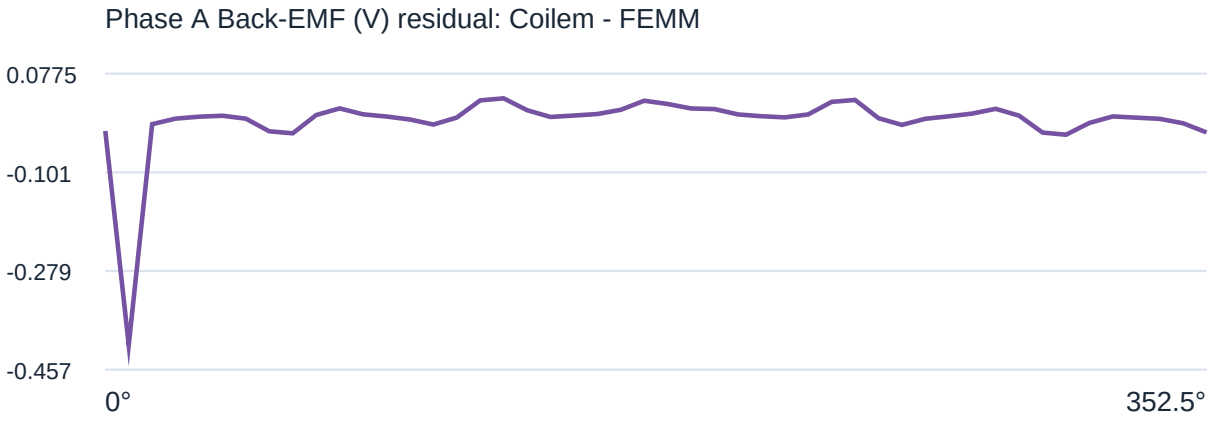
Frozen native fixture JSON



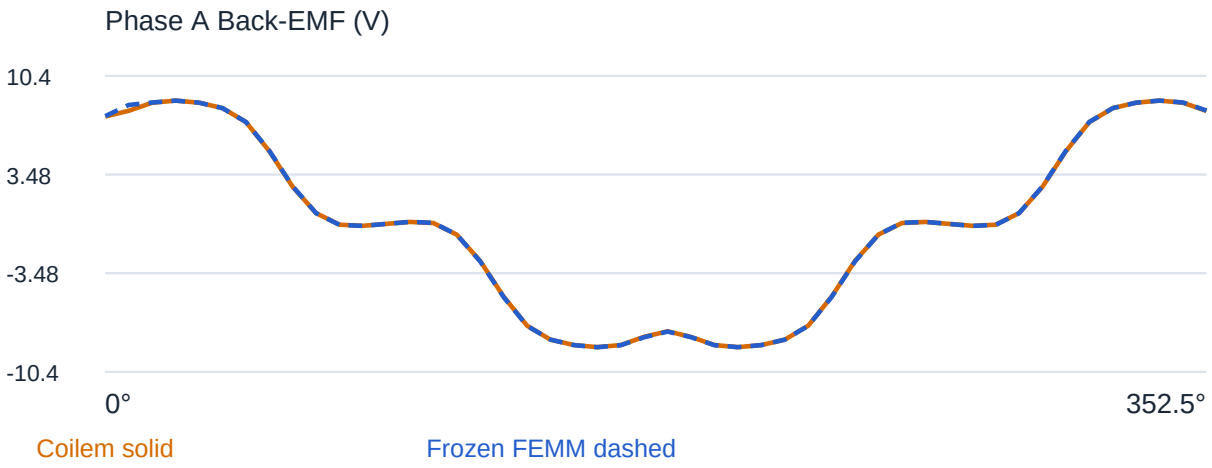

Residual solid




Coilem solid  Frozen FEMM dashed



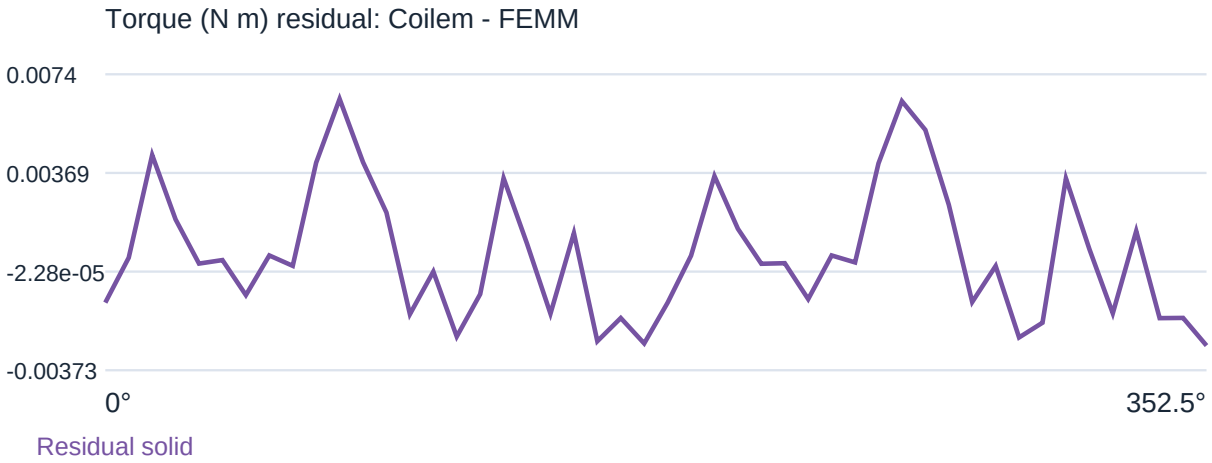

Residual solid



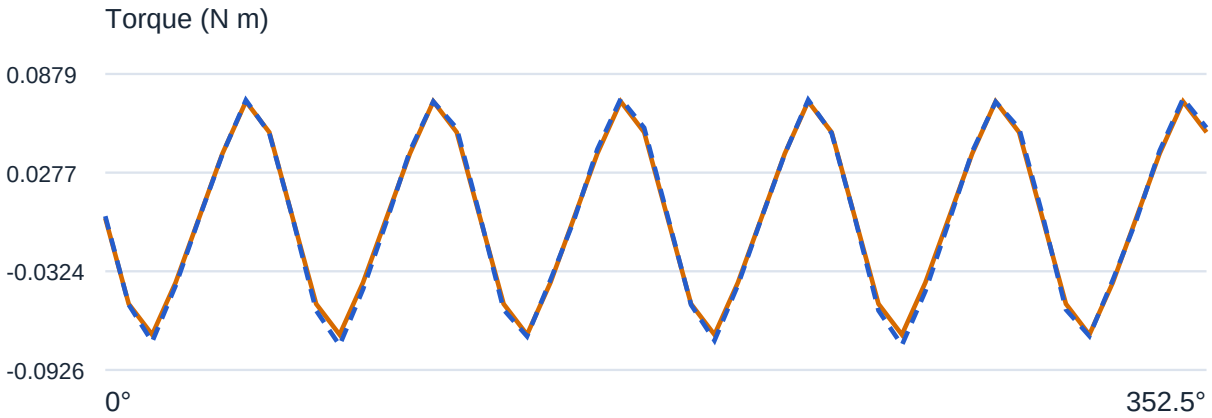

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase A – spm_4p12s_simple_noload__baseline

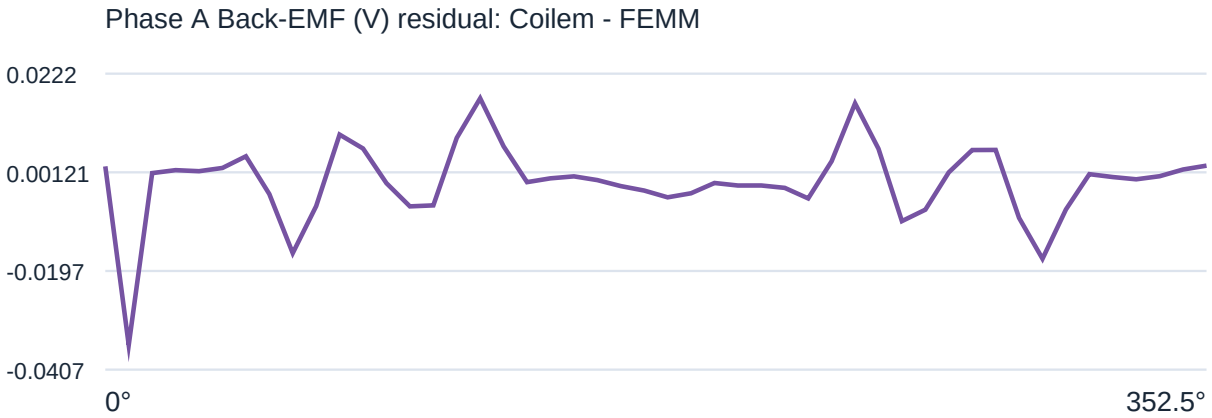
Frozen native fixture JSON



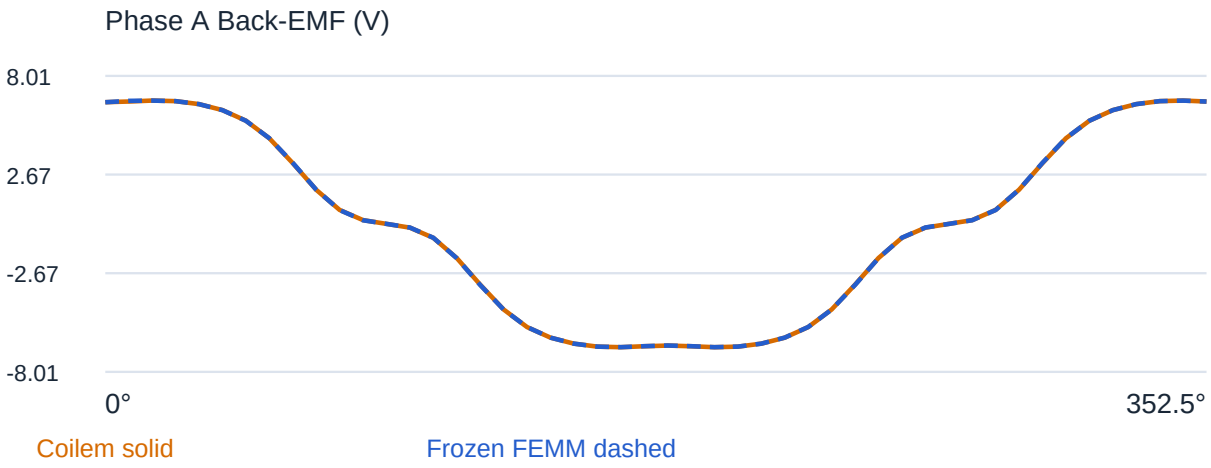

Residual solid




Coilem solid  Frozen FEMM dashed



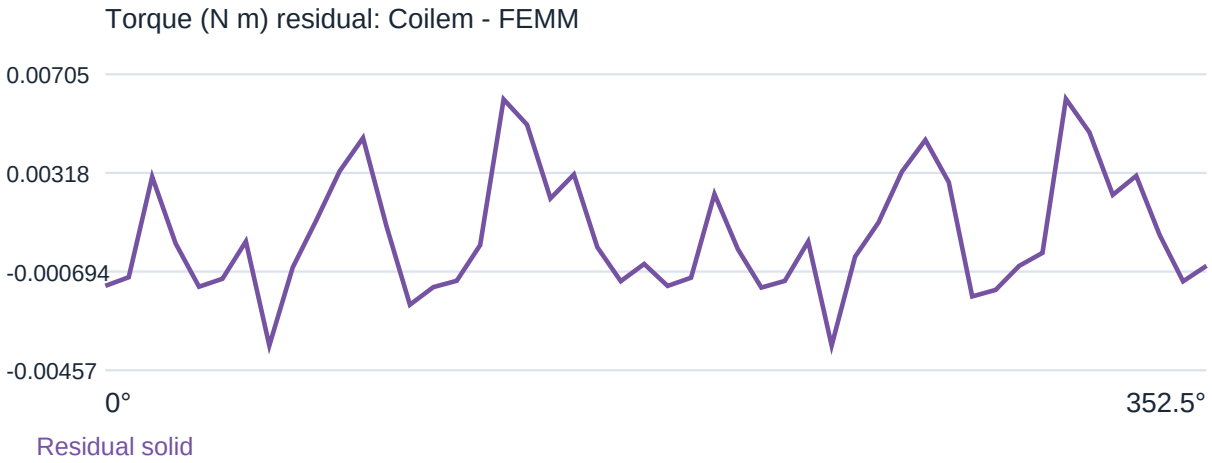

Residual solid



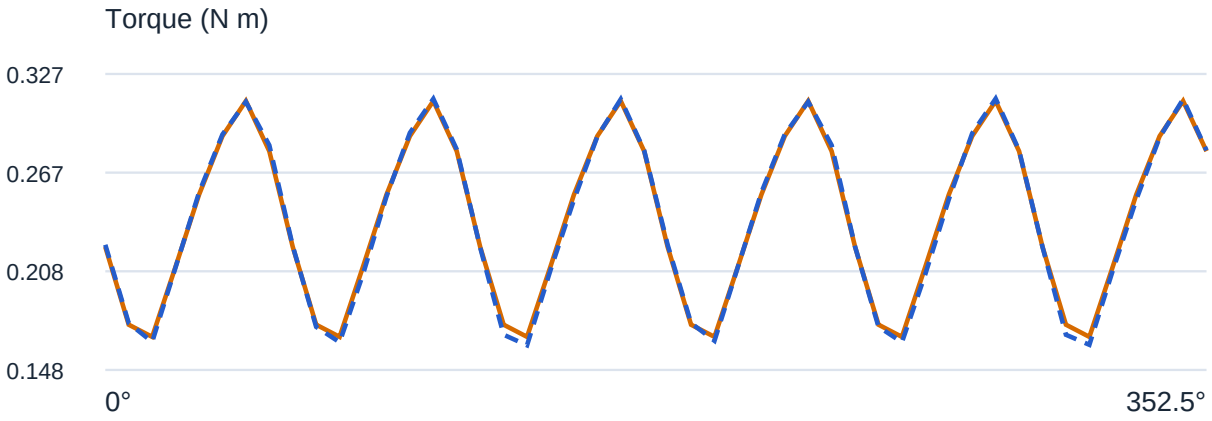

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase A – spm_4p12s_simple_noload__high_current

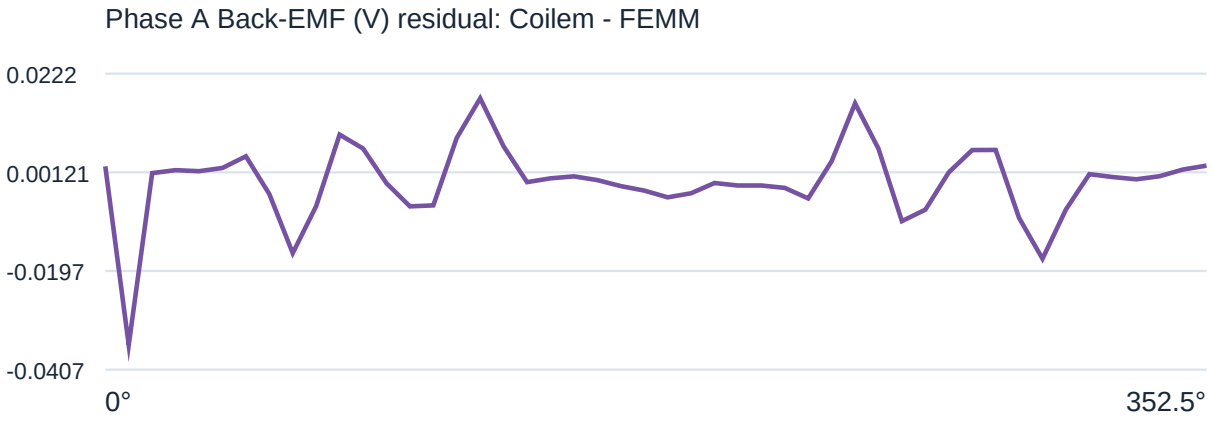
Frozen native fixture JSON



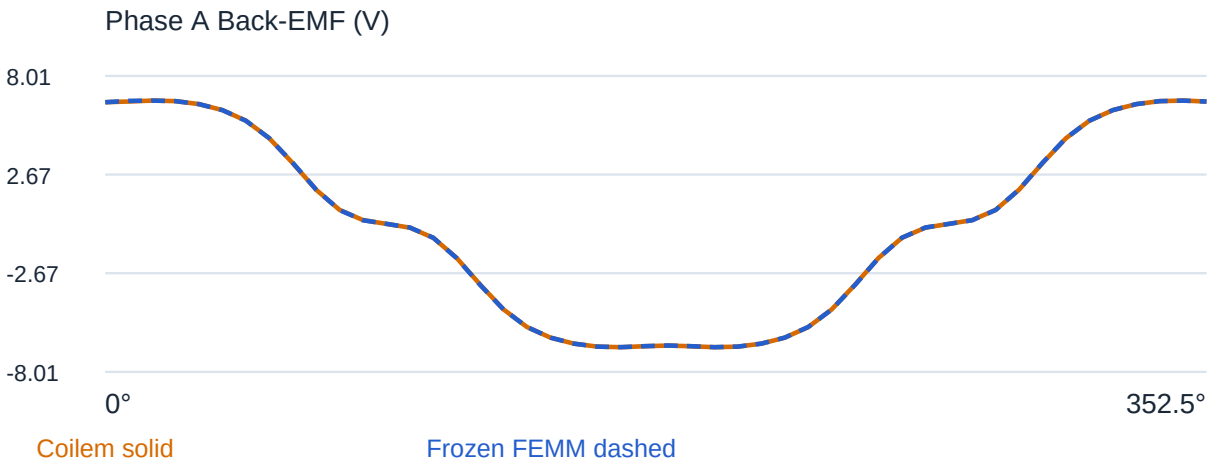

Residual solid




Coilem solid  Frozen FEMM dashed



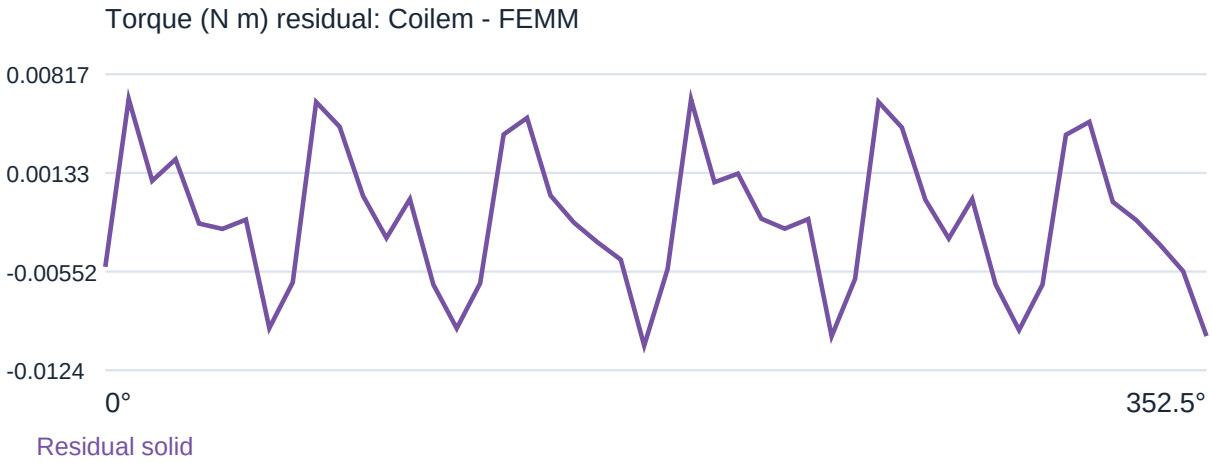

Residual solid



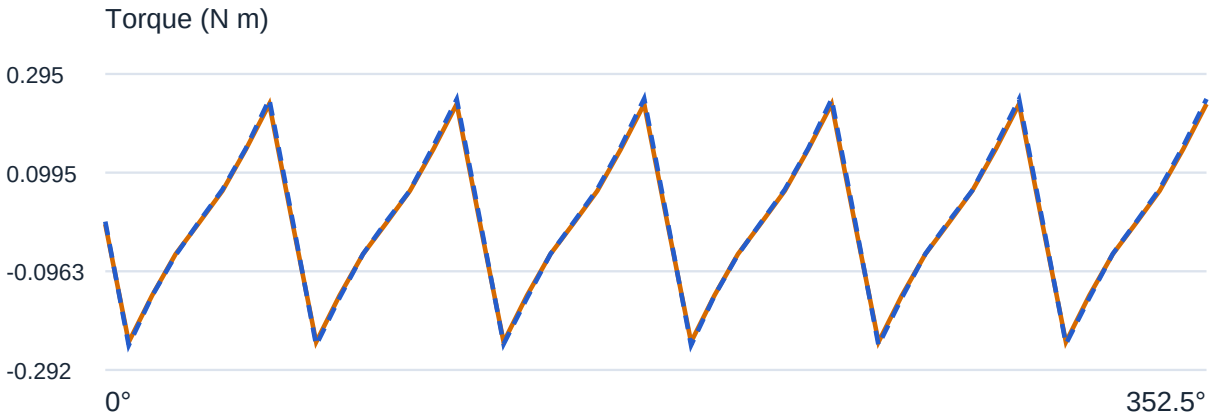

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase A – spm_4p12s_simple_noload__tight_airgap

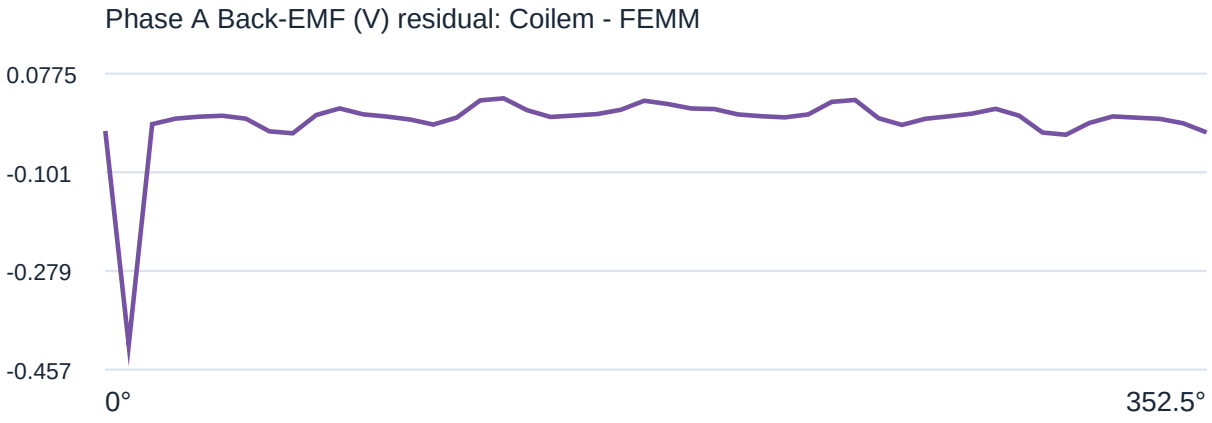
Frozen native fixture JSON



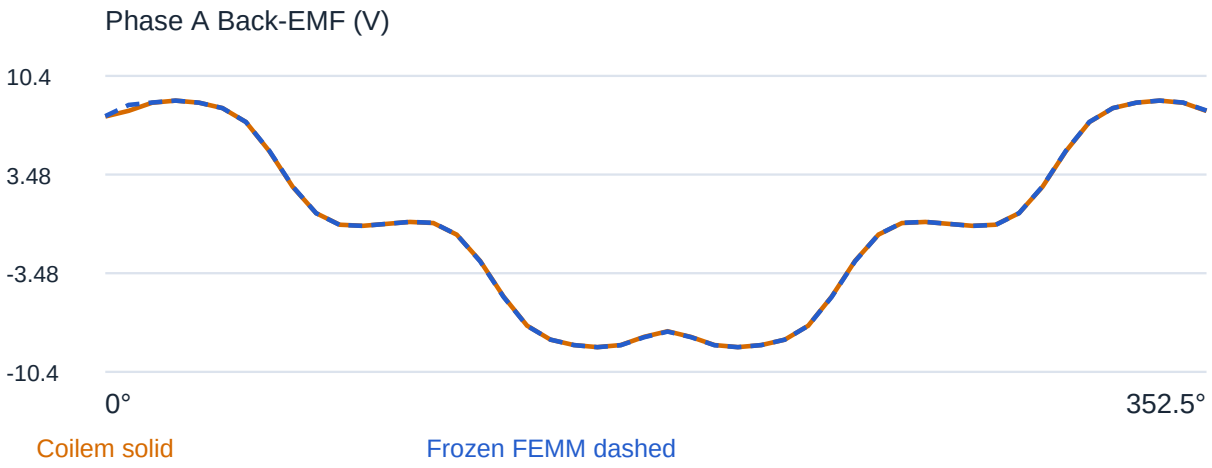

Residual solid




Coilem solid  Frozen FEMM dashed



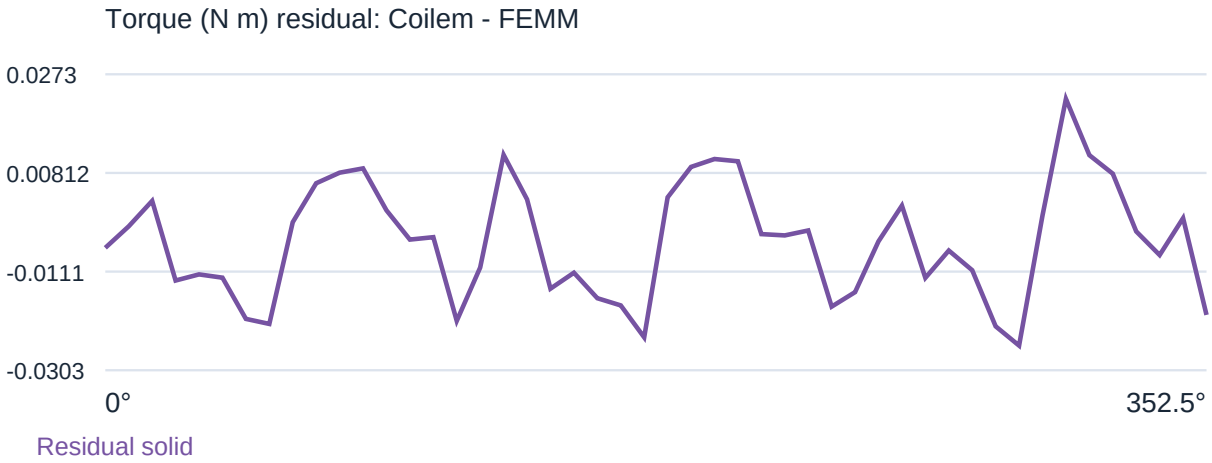

Residual solid



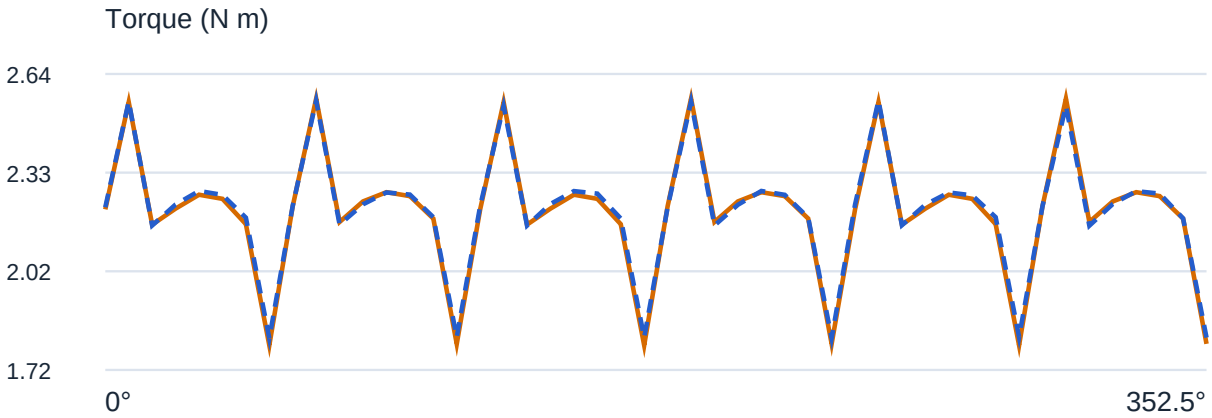

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase A – spm_8p12s_concentrated__baseline

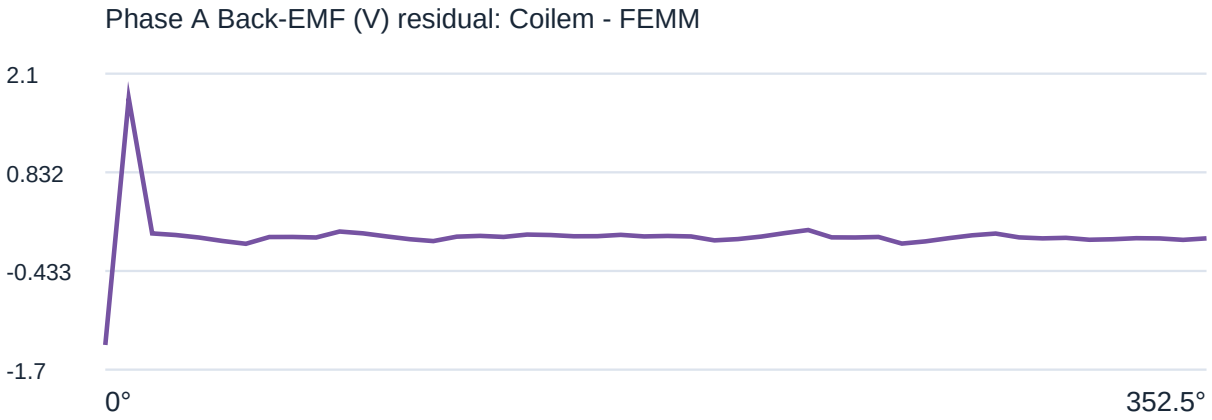
Frozen native fixture JSON



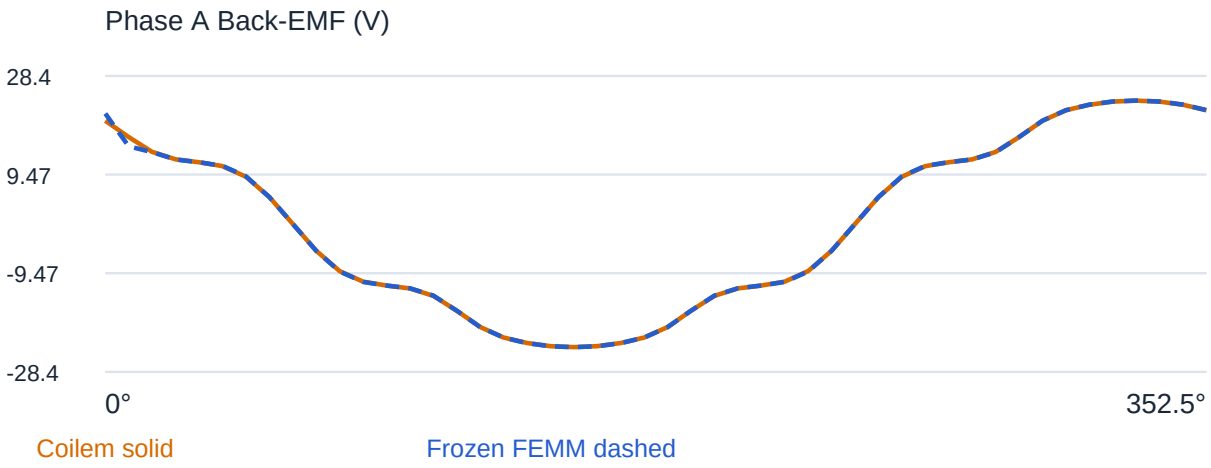

Residual solid




Coilem solid  Frozen FEMM dashed



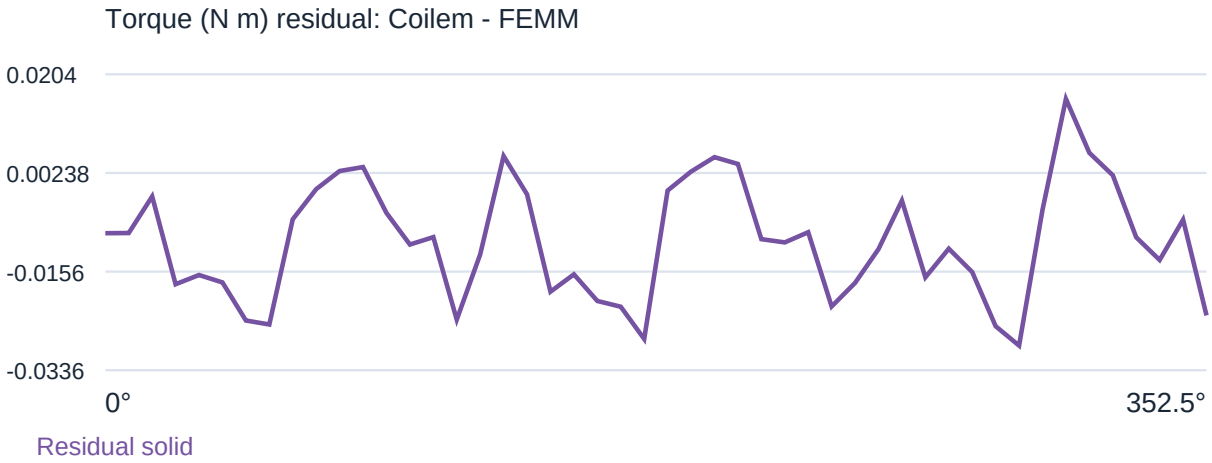

Residual solid

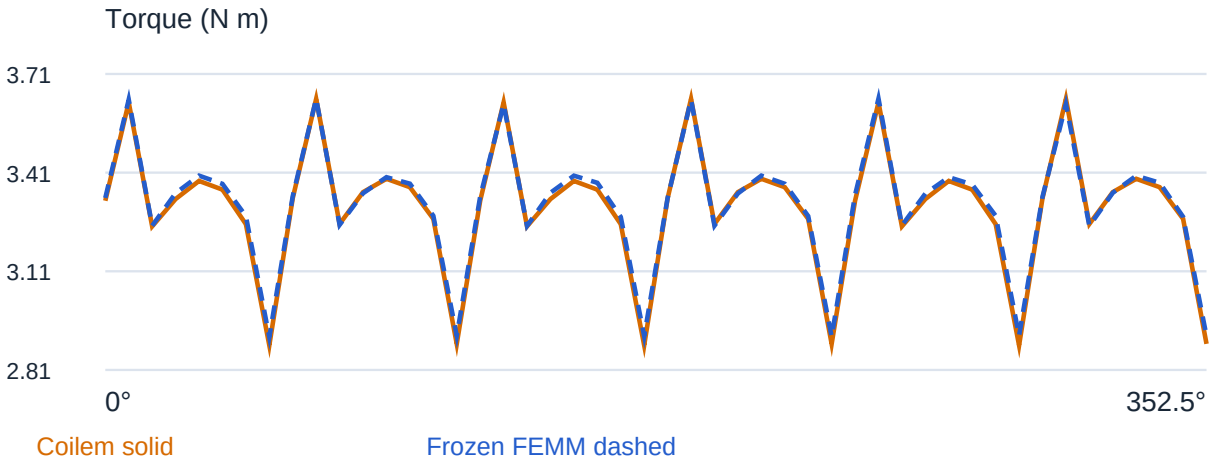


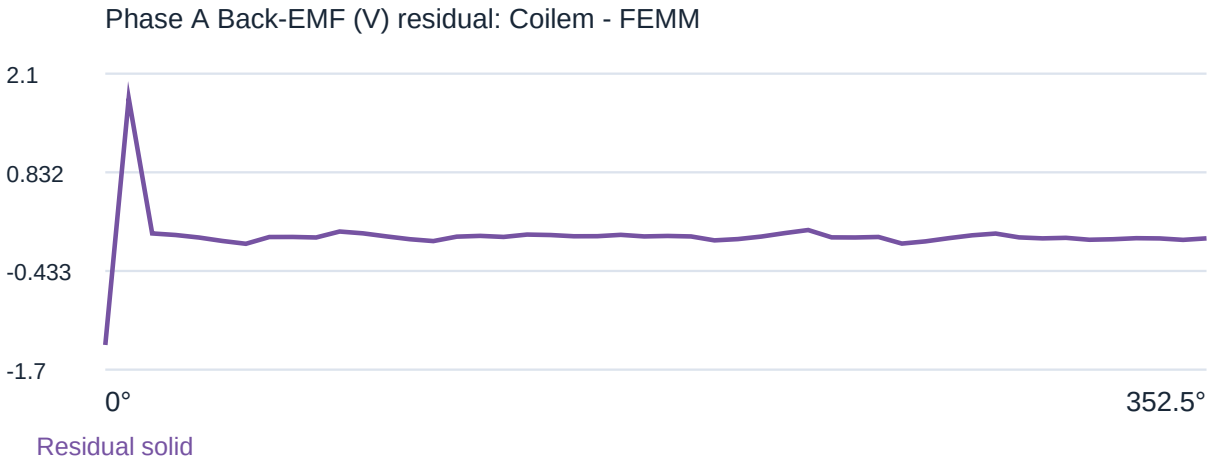

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

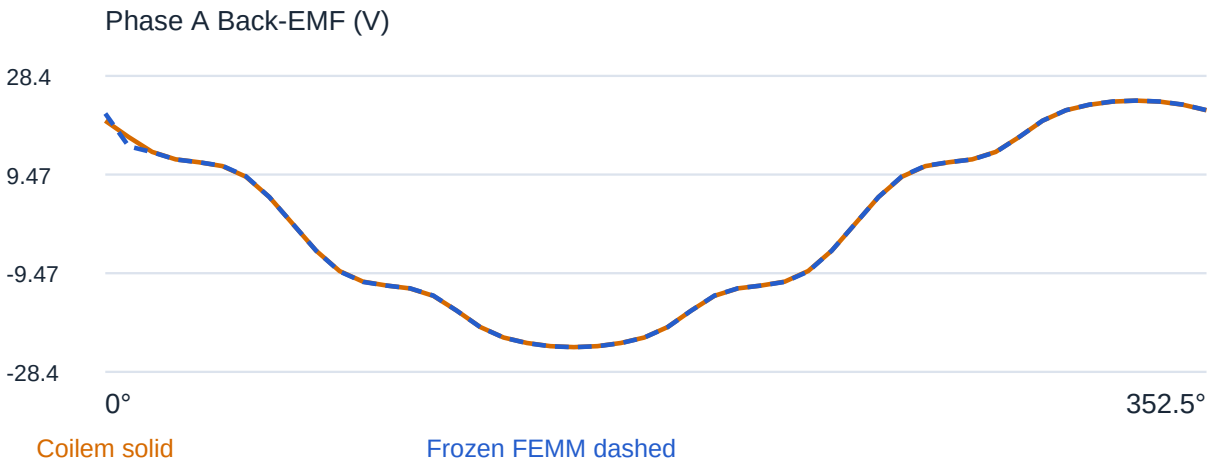
# Phase A – spm_8p12s_concentrated__high_current

Frozen native fixture JSON



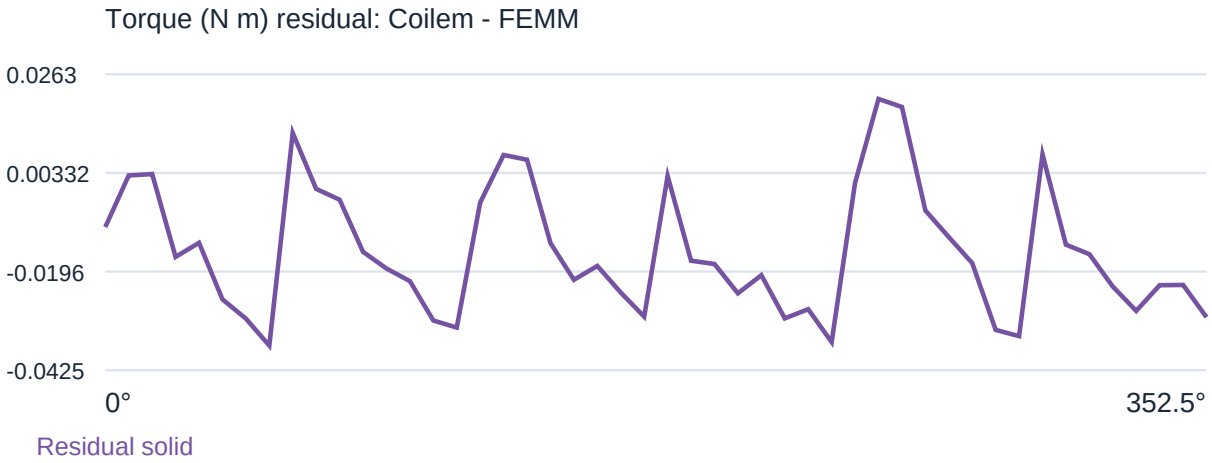



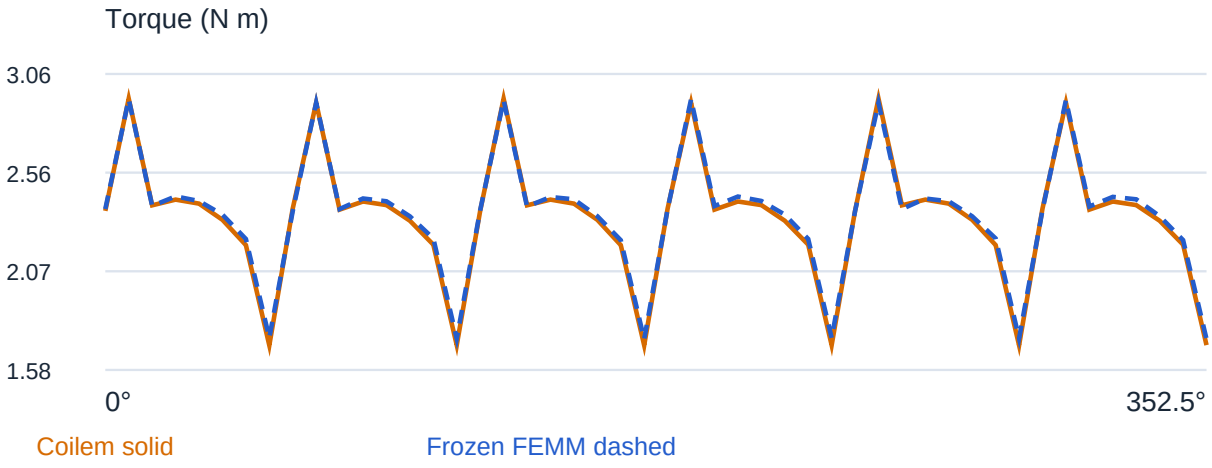


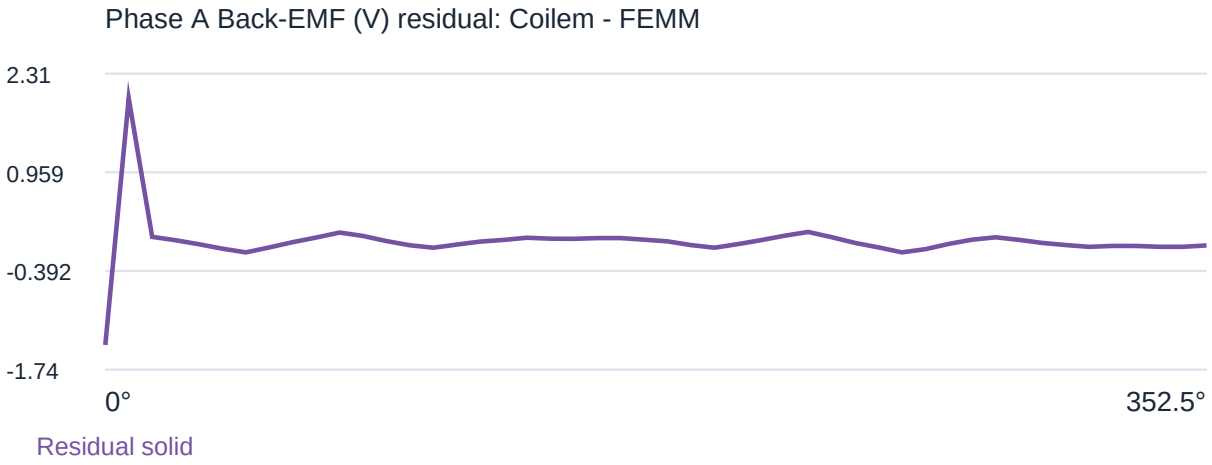

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

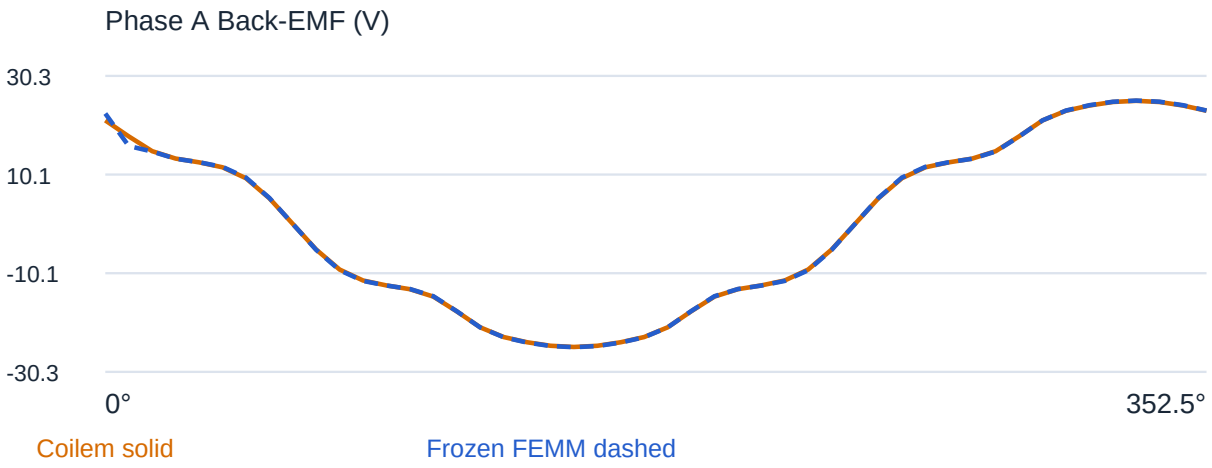
# Phase A – spm_8p12s_concentrated__tight_airgap

Frozen native fixture JSON



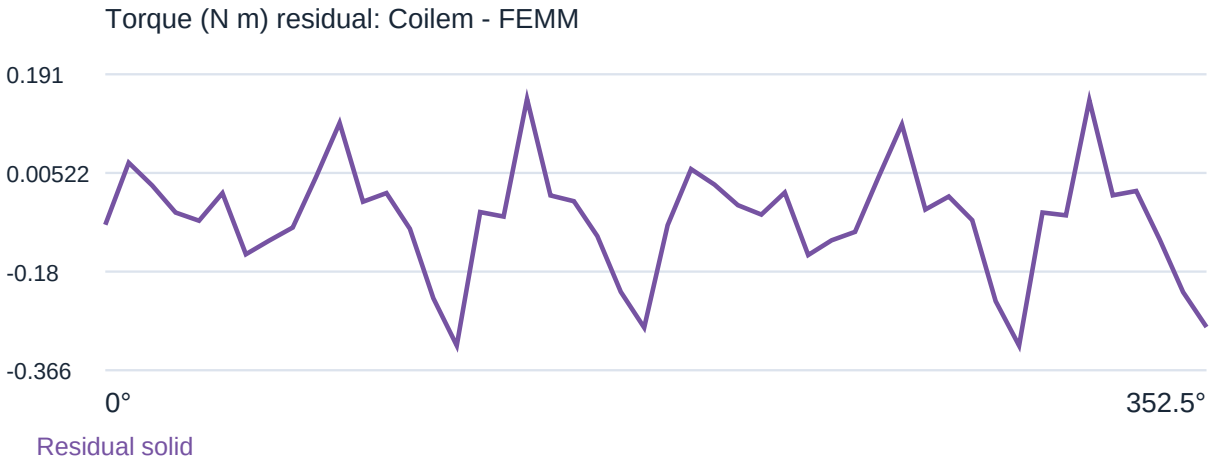





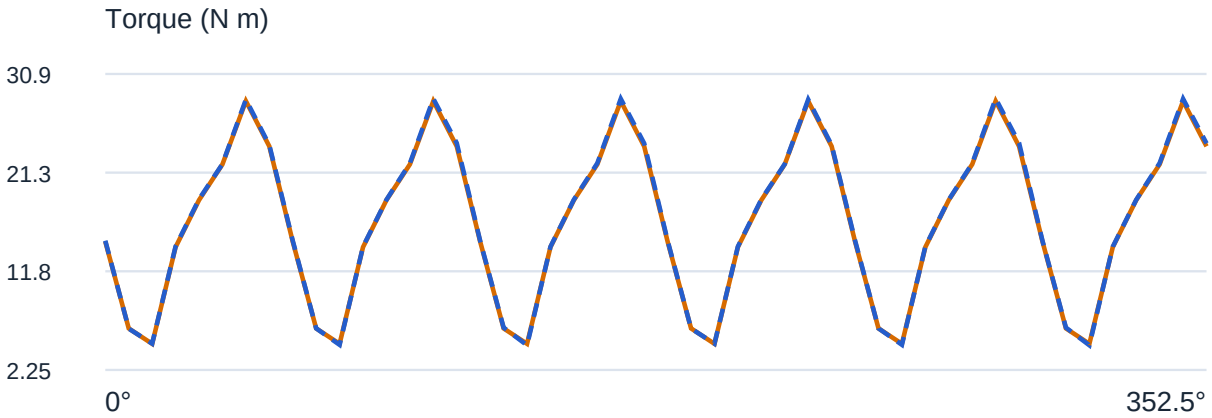

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase A – spm_8p12s_reference_angle30__baseline

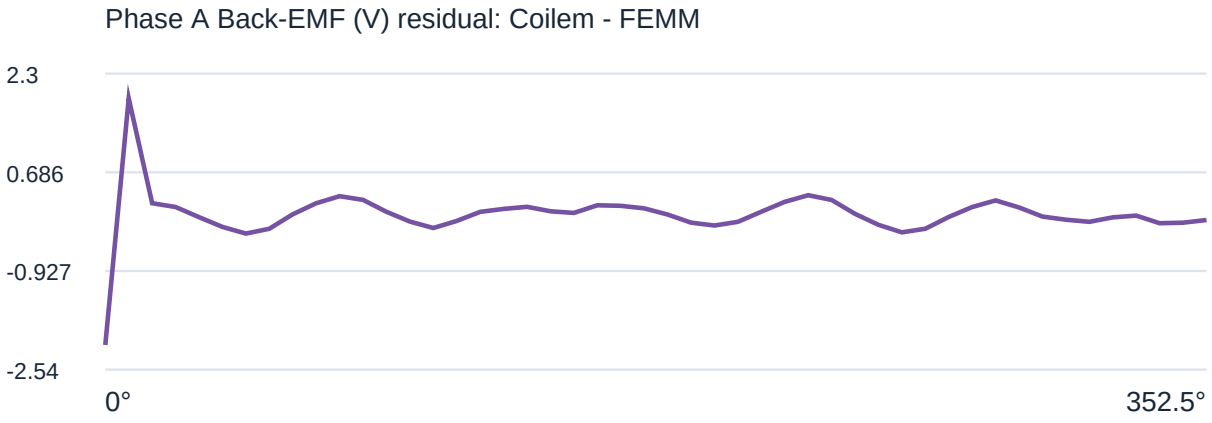
Frozen native fixture JSON



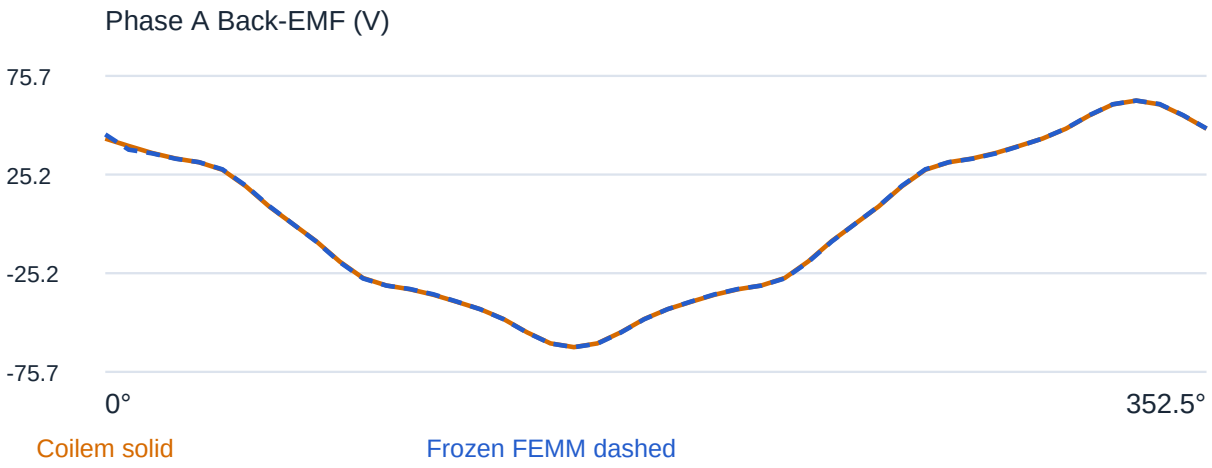

Residual solid




Coilem solid  Frozen FEMM dashed



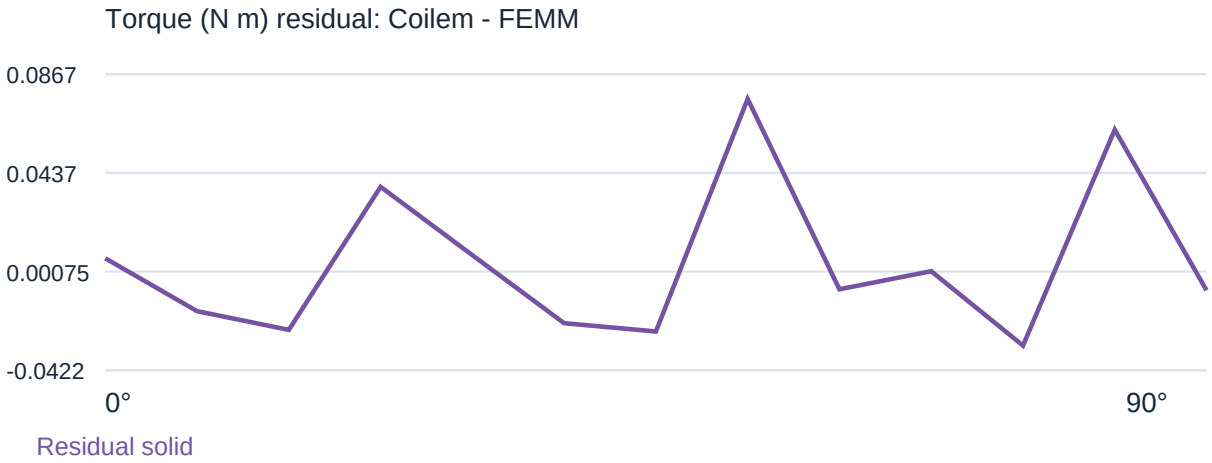

Residual solid



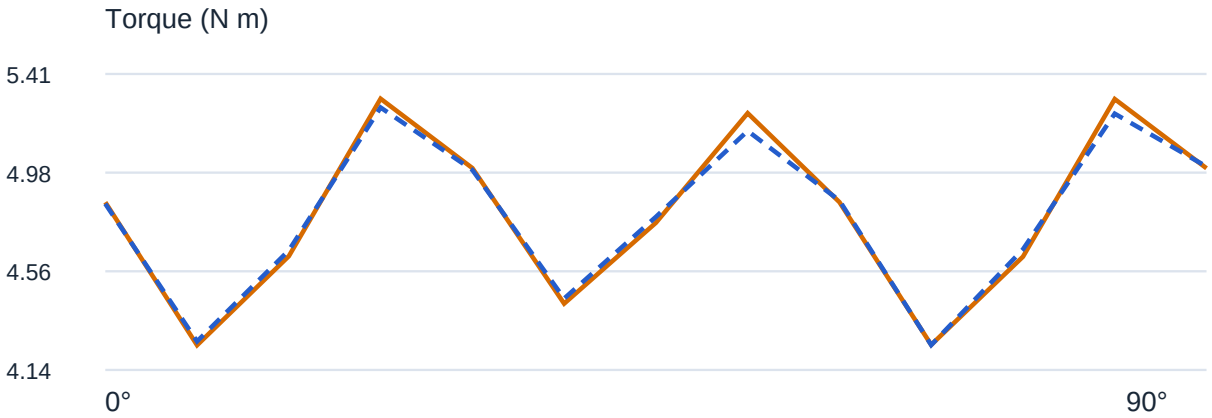

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase B – ipm_flat_12s10p_fscw_contract

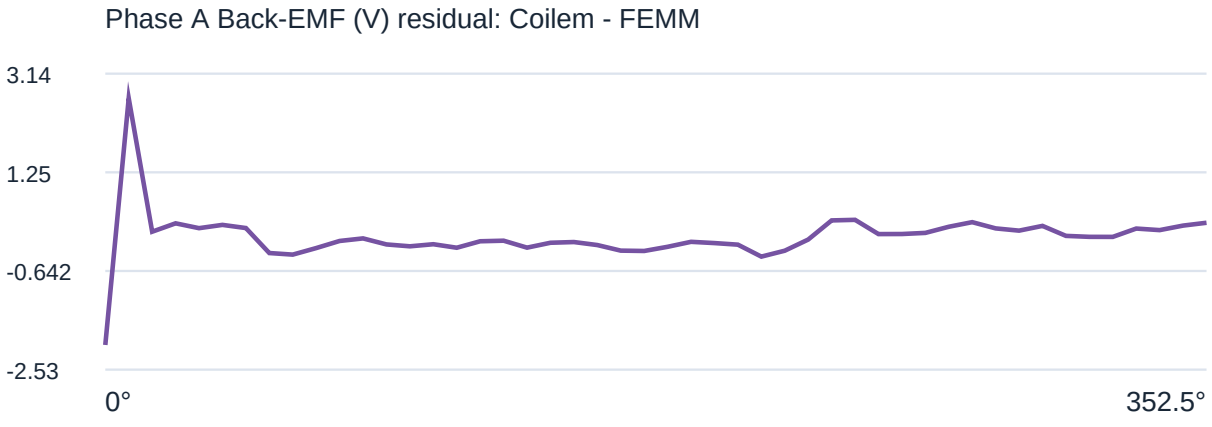
Frozen native fixture JSON



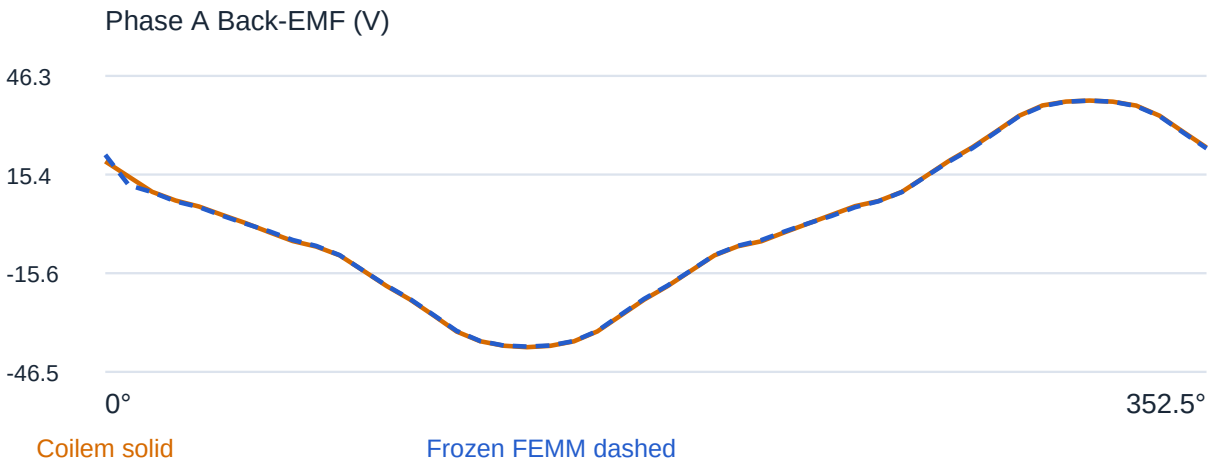

Residual solid




Coilem solid  Frozen FEMM dashed



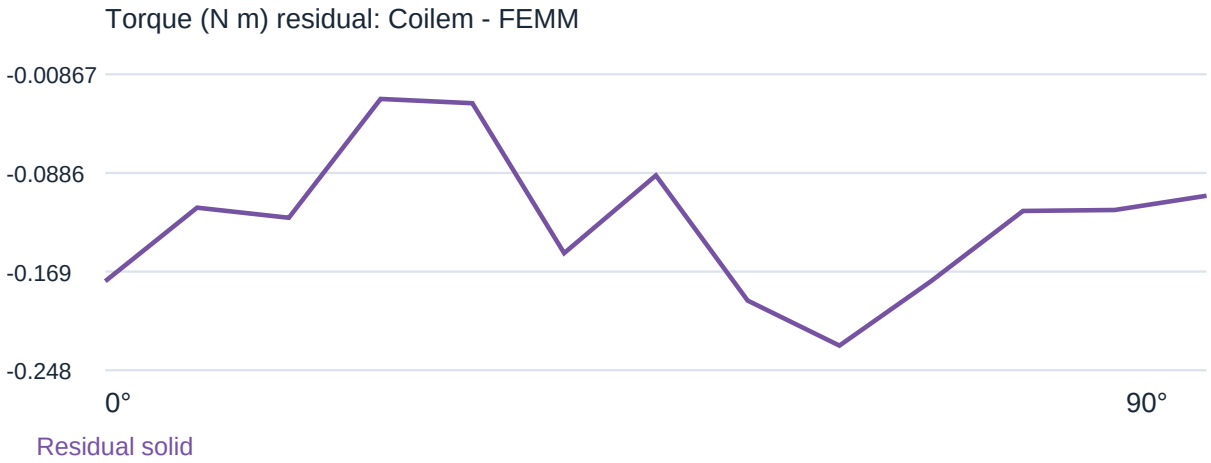

Residual solid



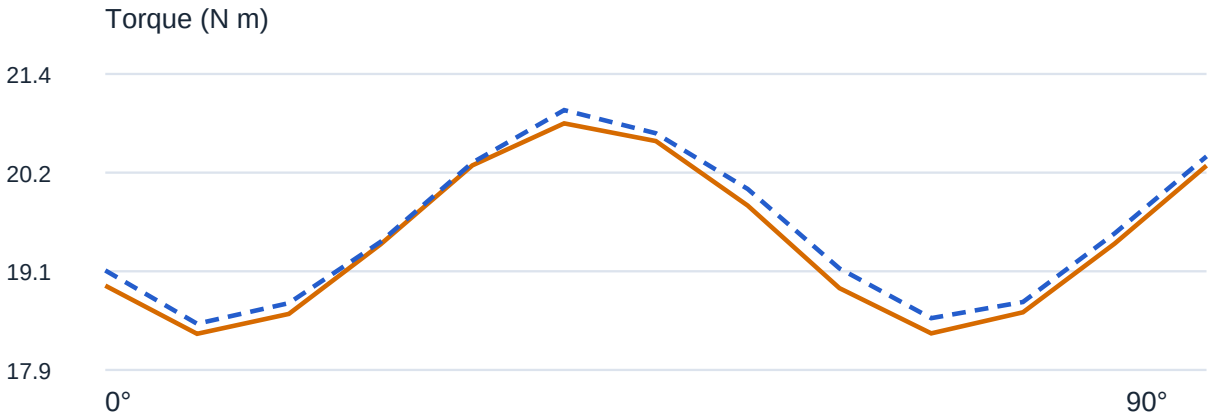

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

## Phase B – ipm_flat_24s16p_industrial_contract

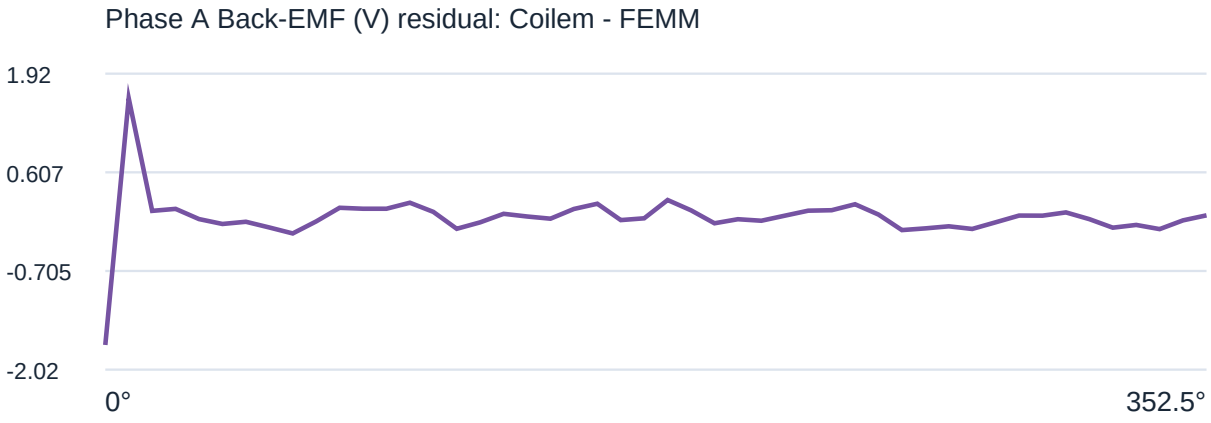
Frozen native fixture JSON



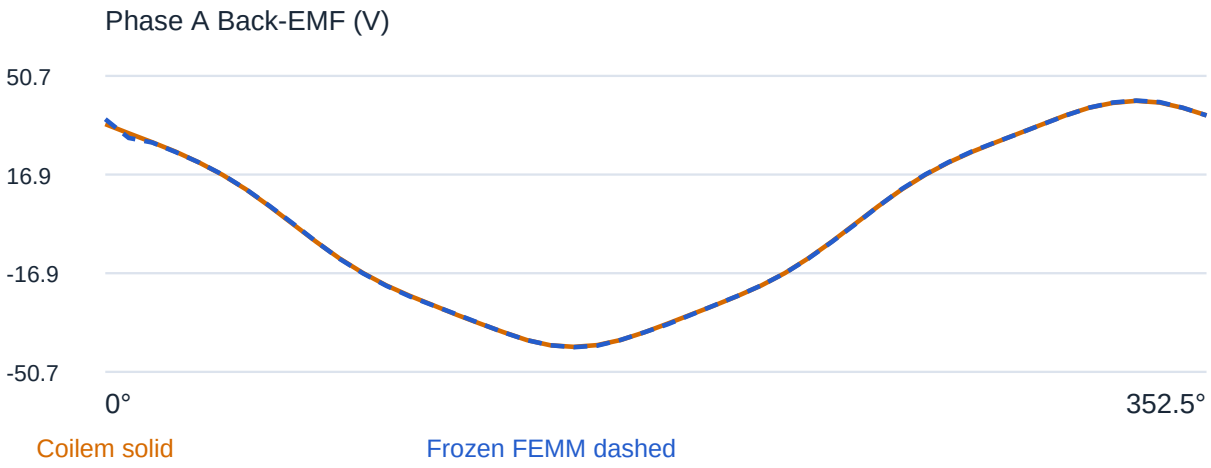

Residual solid




Coilem solid  Frozen FEMM dashed



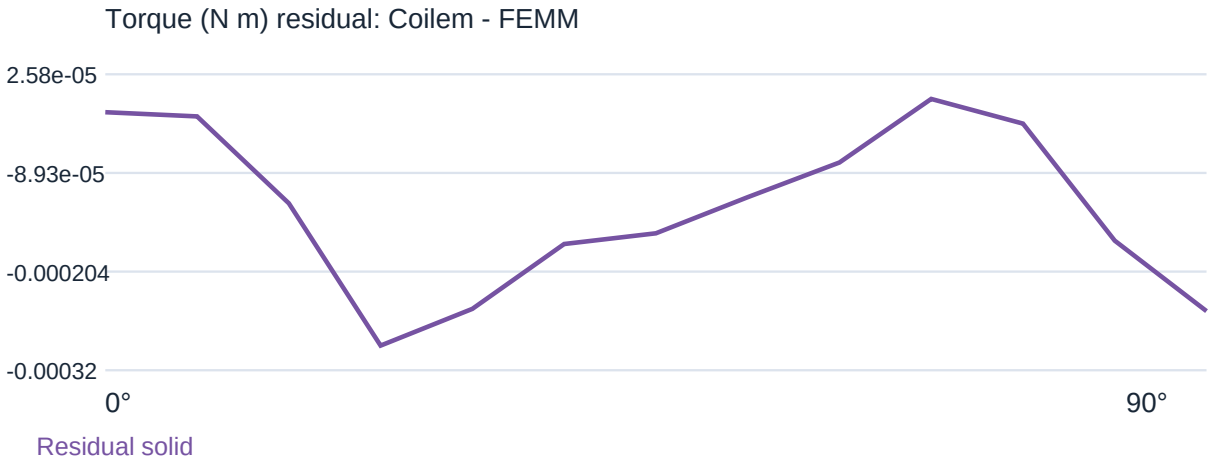

Residual solid



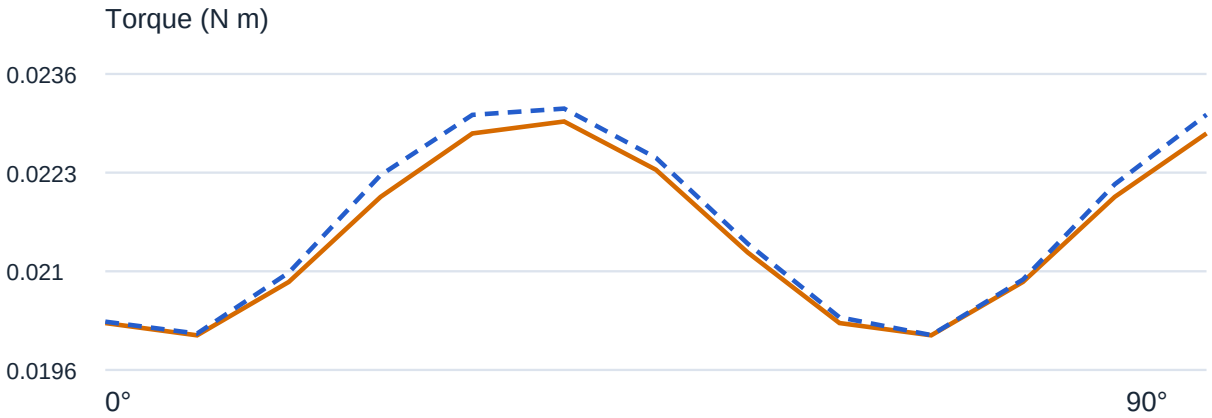

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

## Phase B – ipm_flat_4p12s_micro_contract

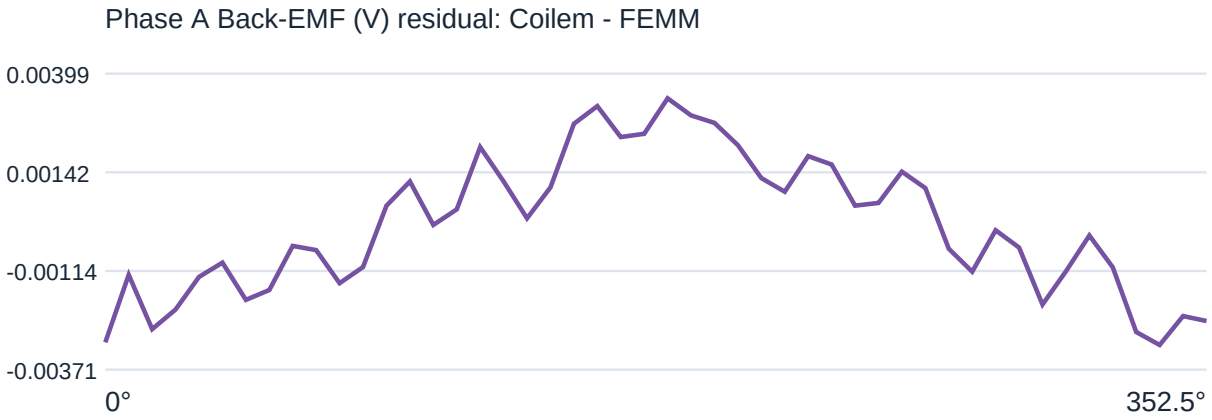
Frozen native fixture JSON



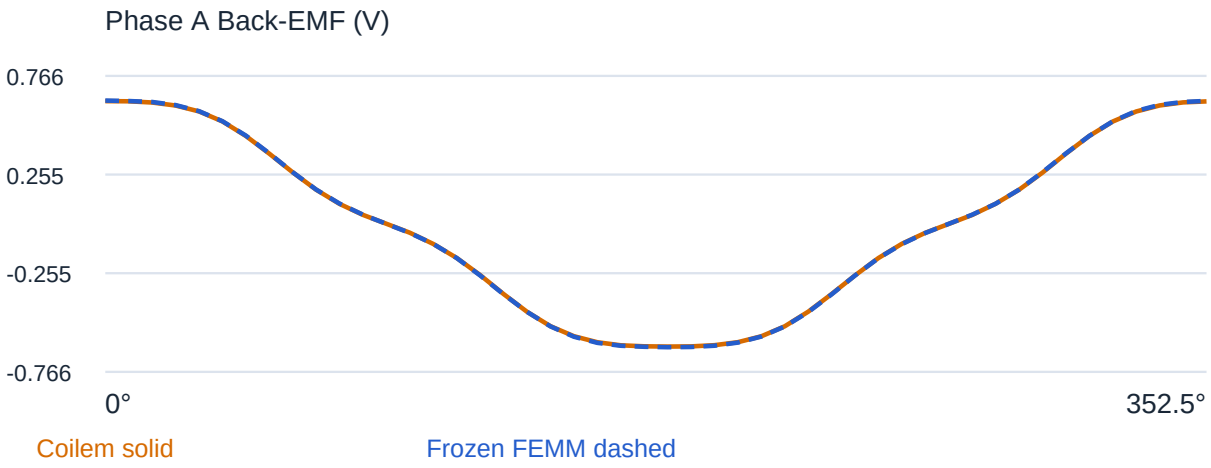

Residual solid




Coilem solid  Frozen FEMM dashed



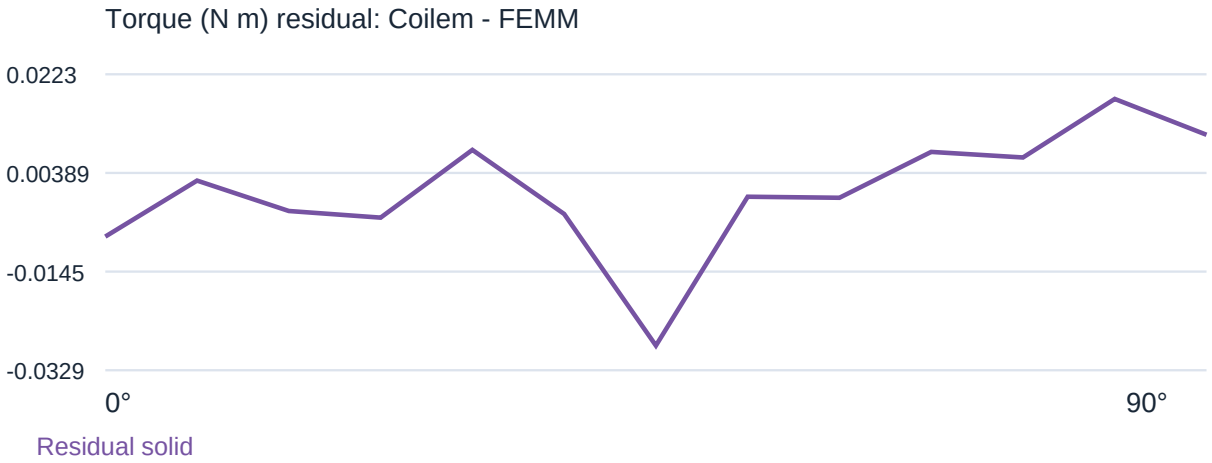

Residual solid

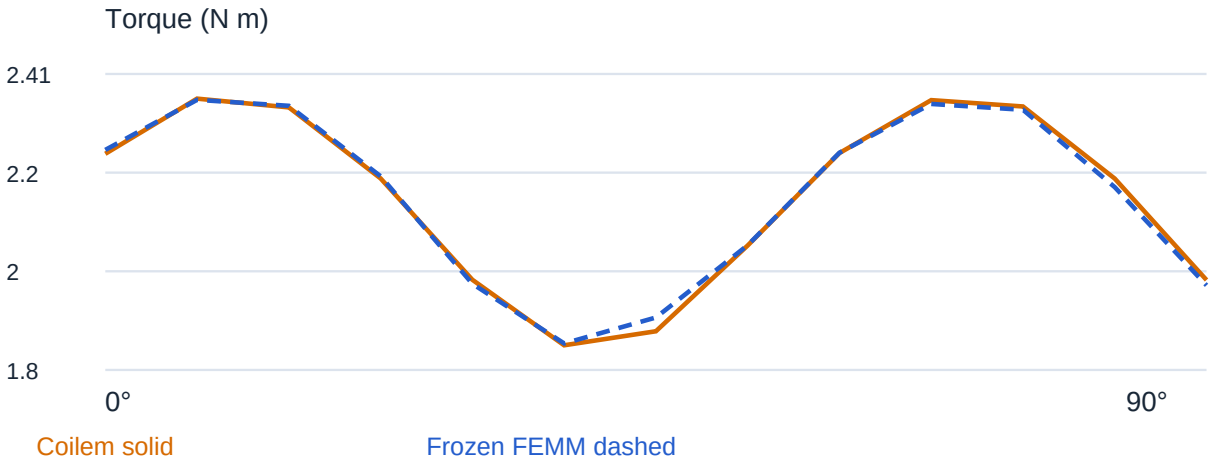


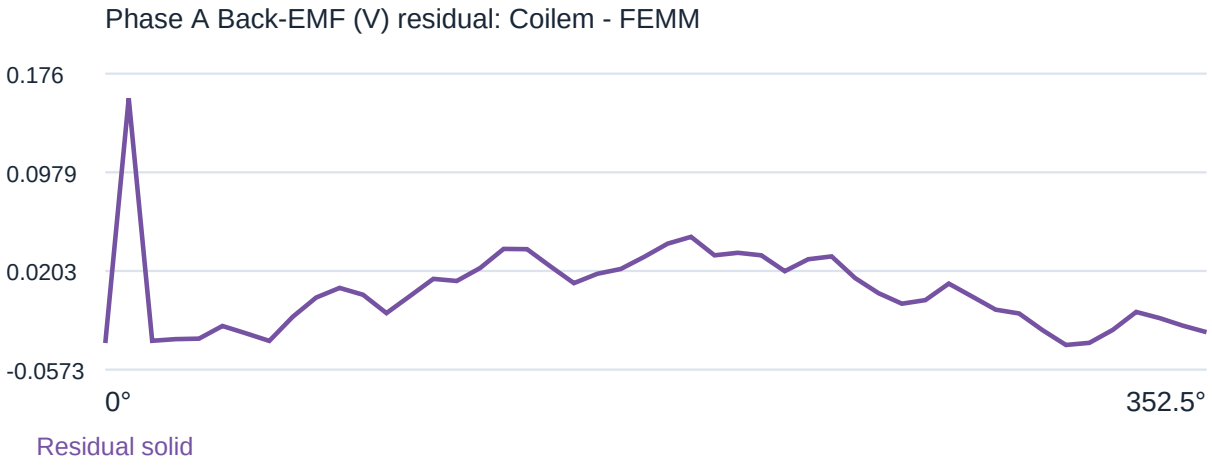

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

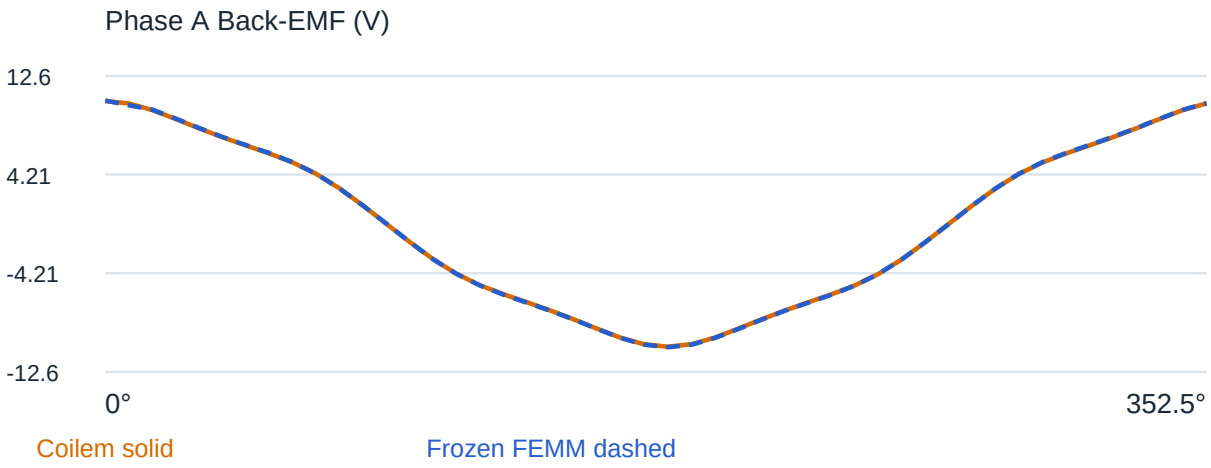
## Phase B – ipm_flat_8p12s_fsae_angled_pocket_contract

Frozen native fixture JSON



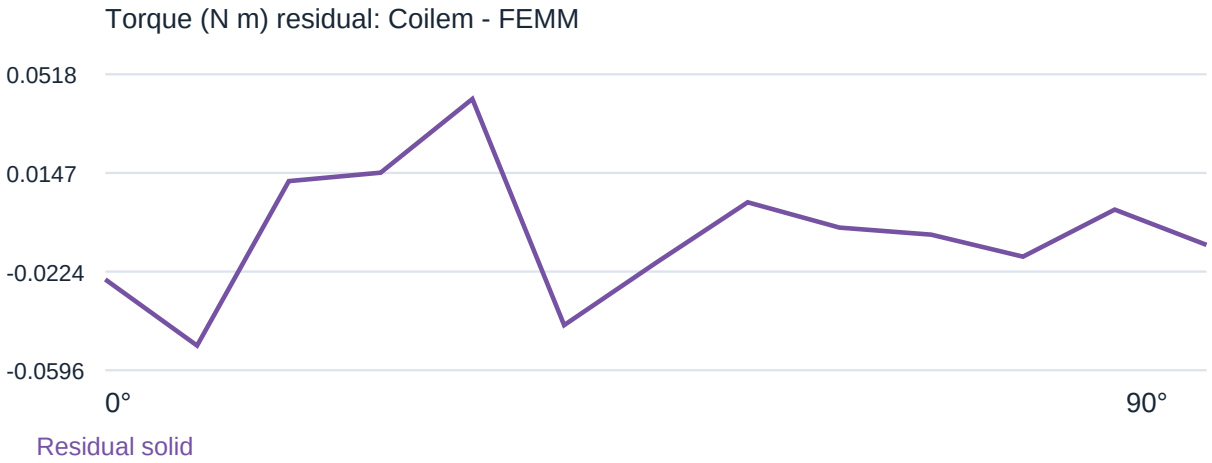





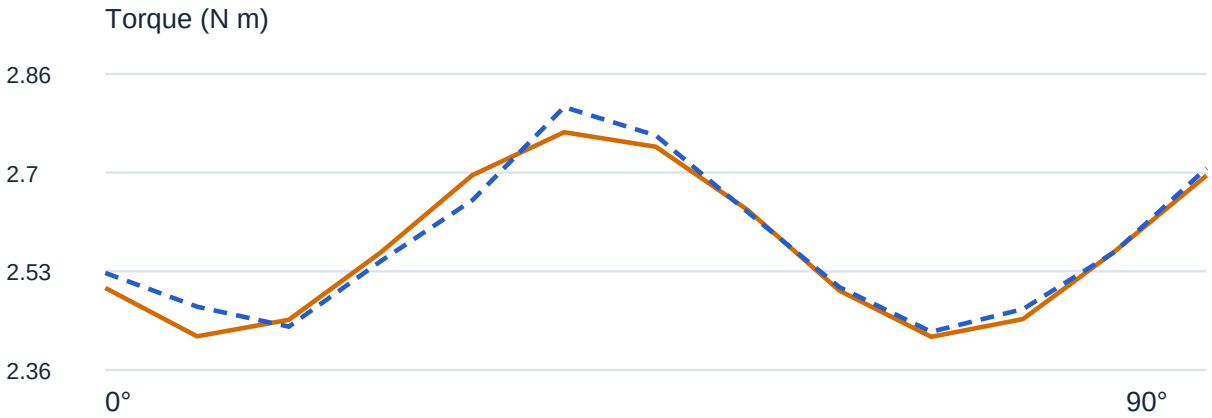

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

## Phase B – ipm_flat_8p12s_fsae_contract

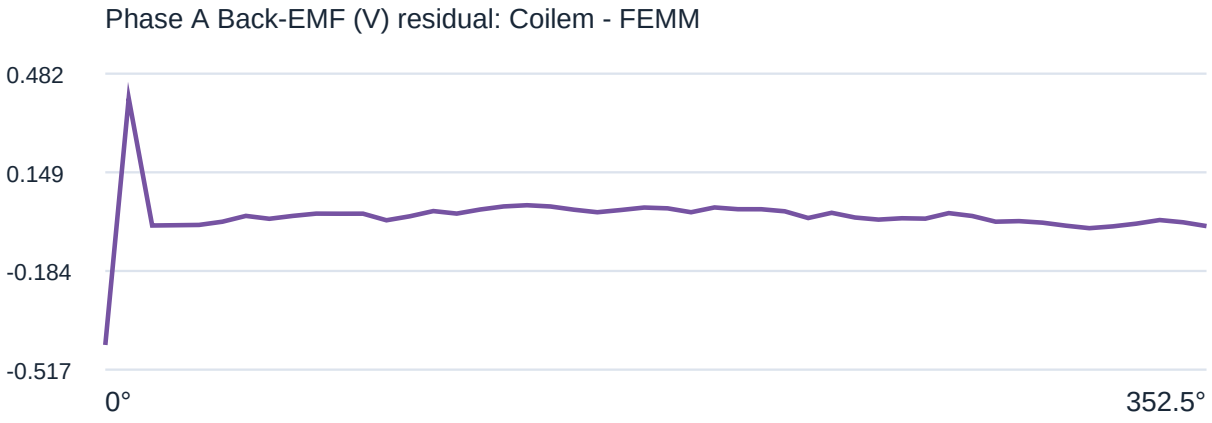
Frozen native fixture JSON



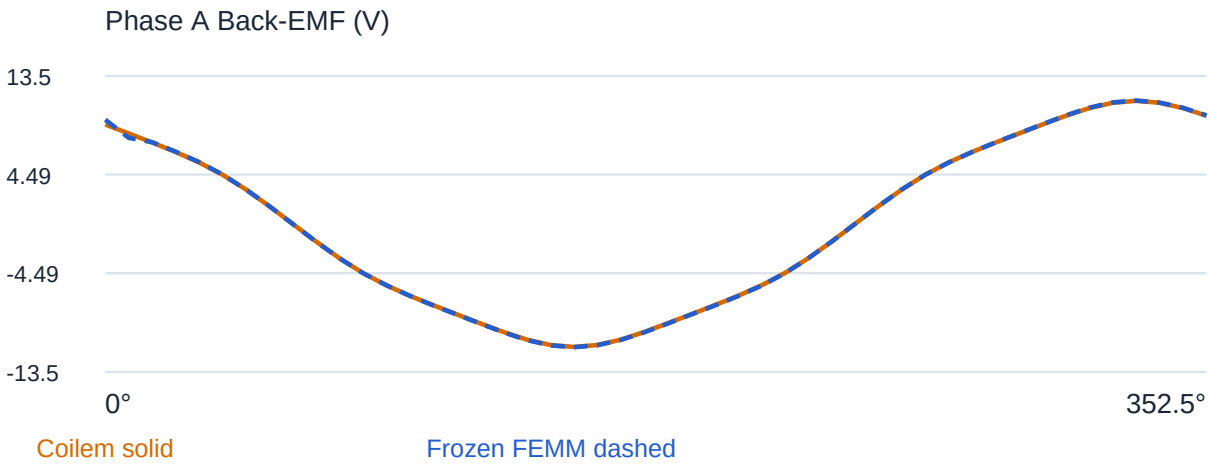

Residual solid




Coilem solid  Frozen FEMM dashed



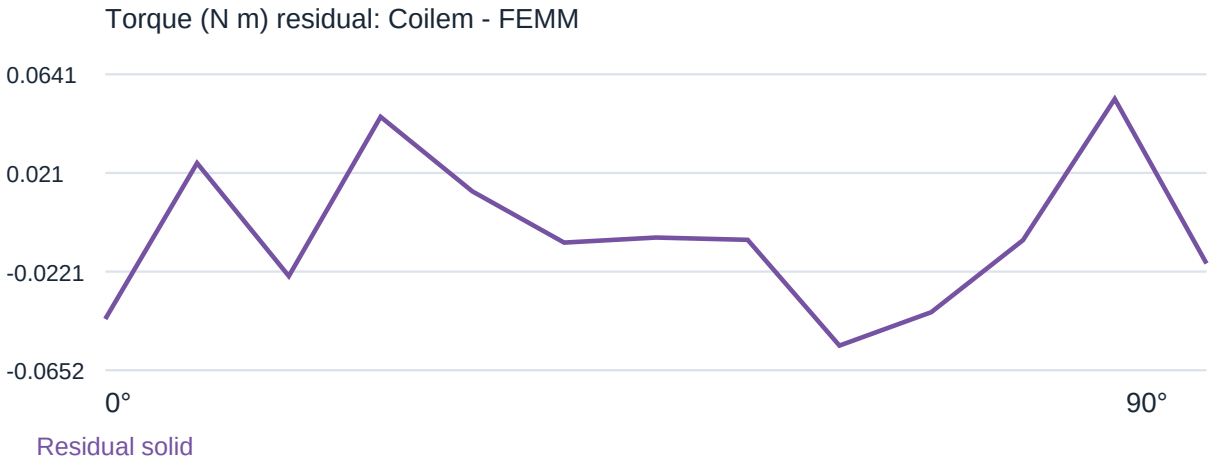

Residual solid



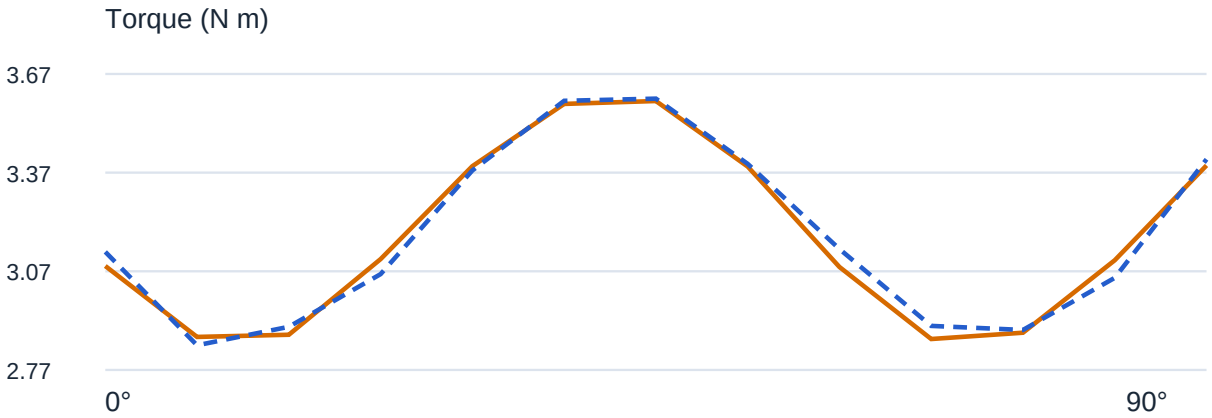

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase B – ipm_flat_8p12s_fsae_thin_bridge_contract

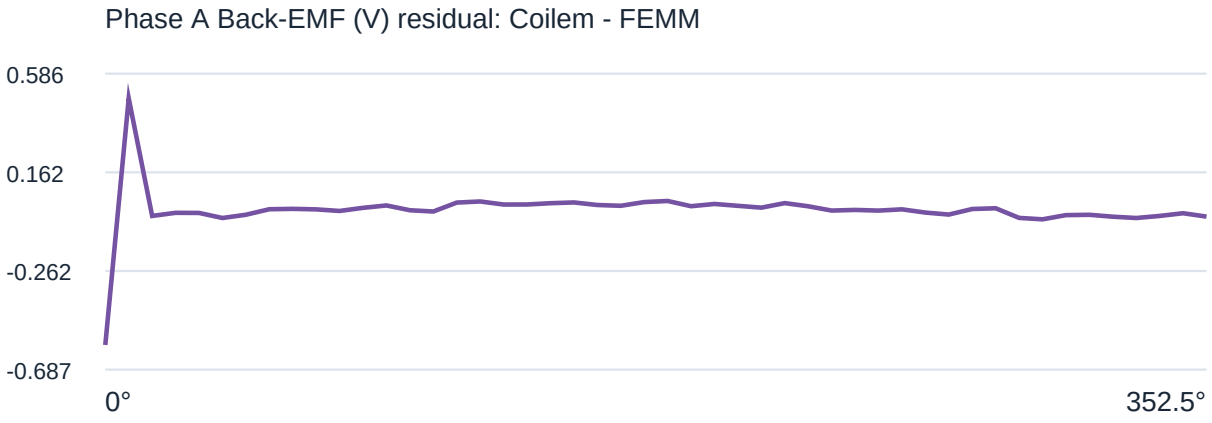
Frozen native fixture JSON



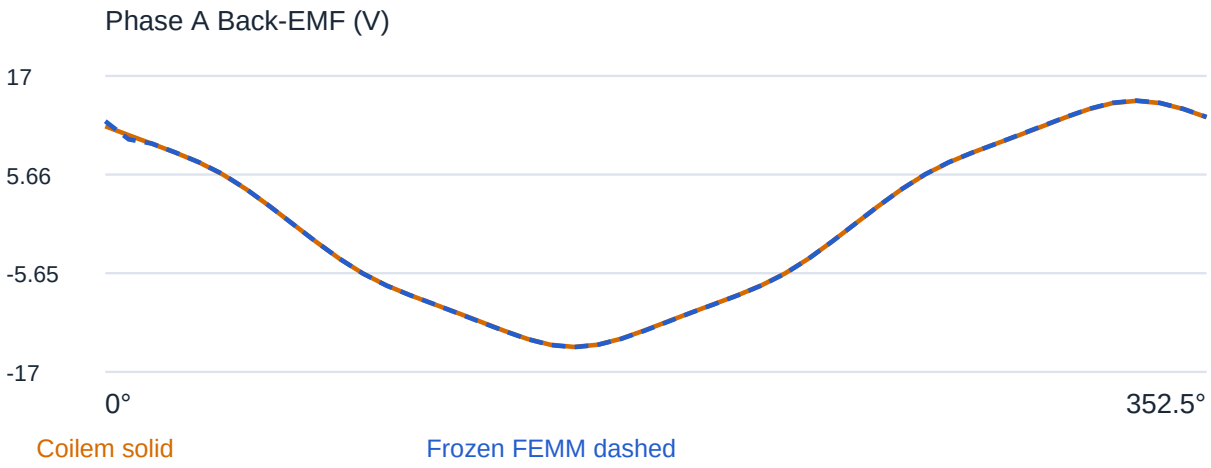

Residual solid




Coilem solid  Frozen FEMM dashed



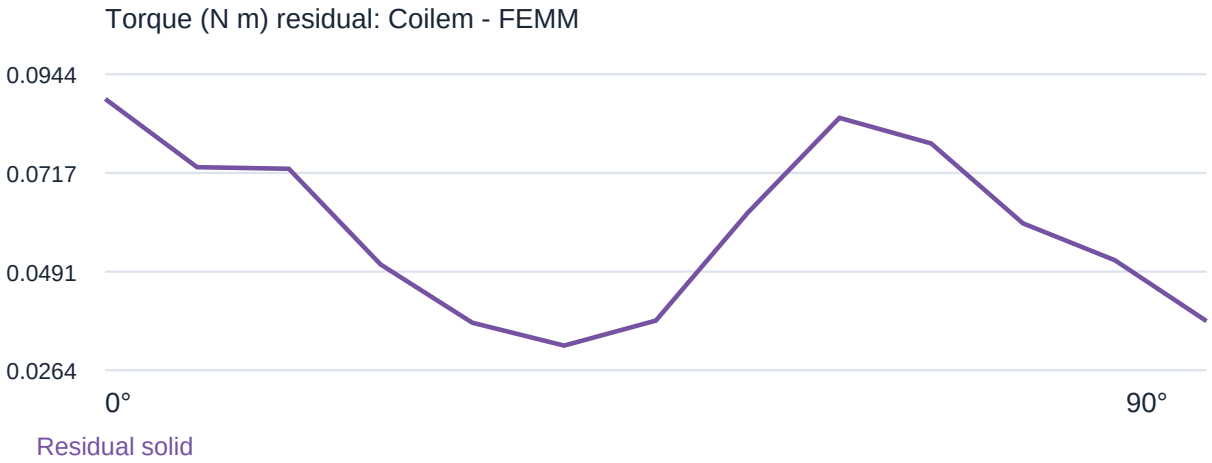

Residual solid



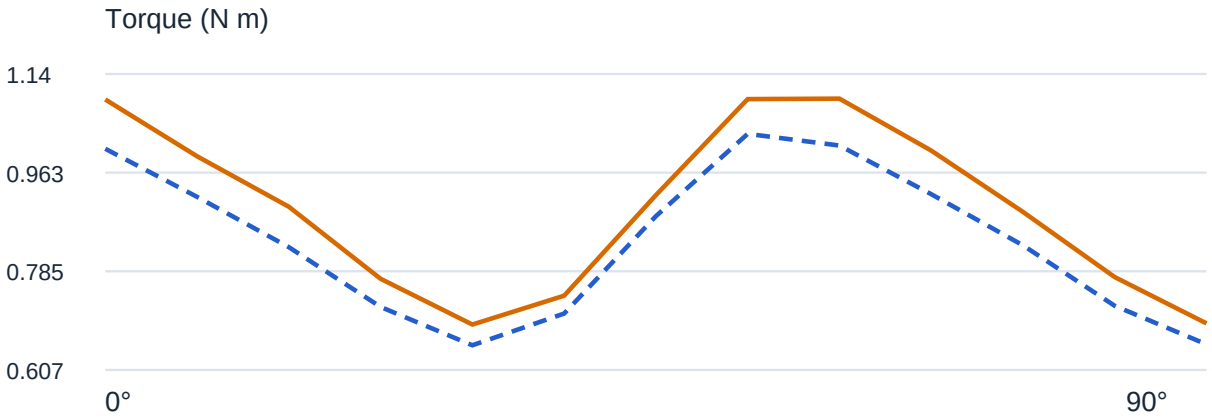

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase B – ipm_vshape_8p12s_baseline_contract

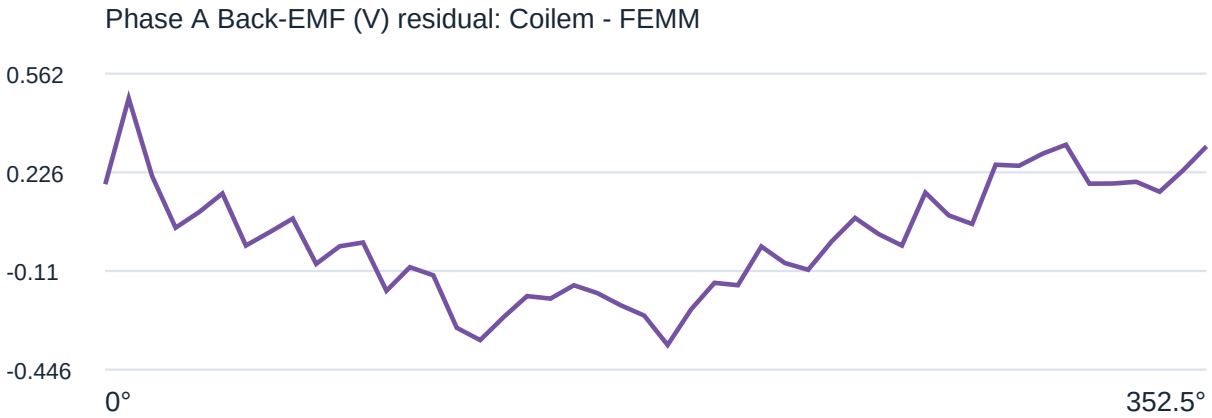
Frozen native fixture JSON



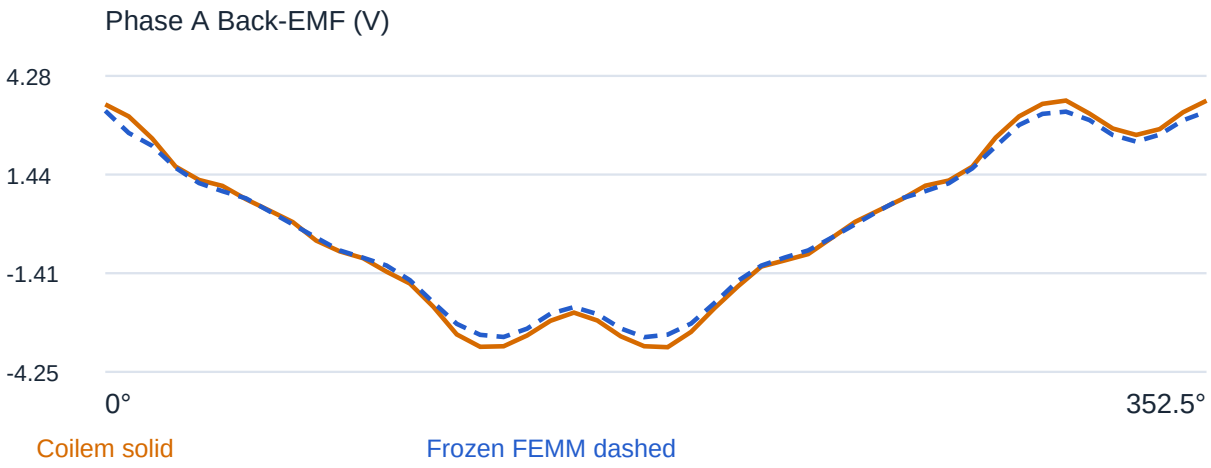

Residual solid




Coilem solid  Frozen FEMM dashed



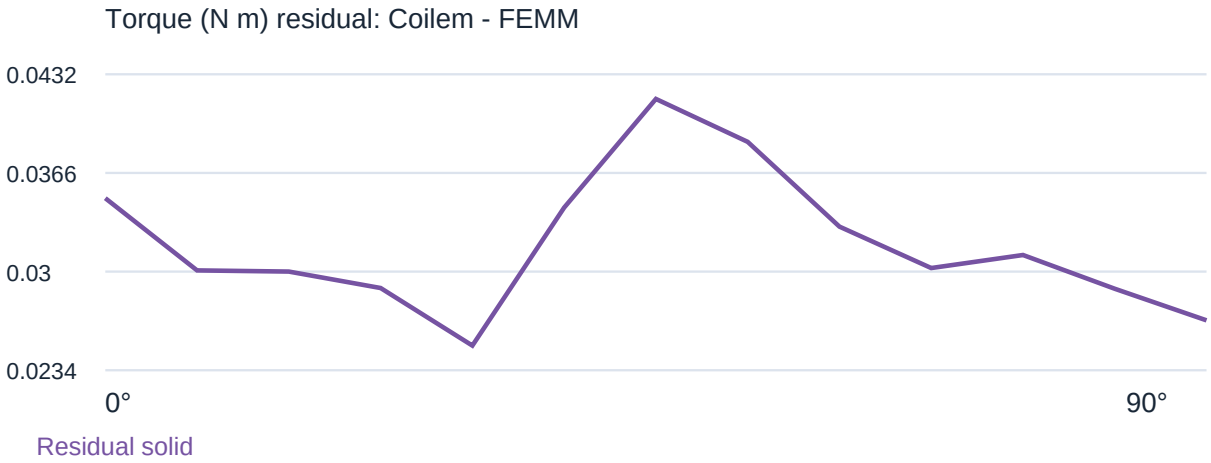

Residual solid

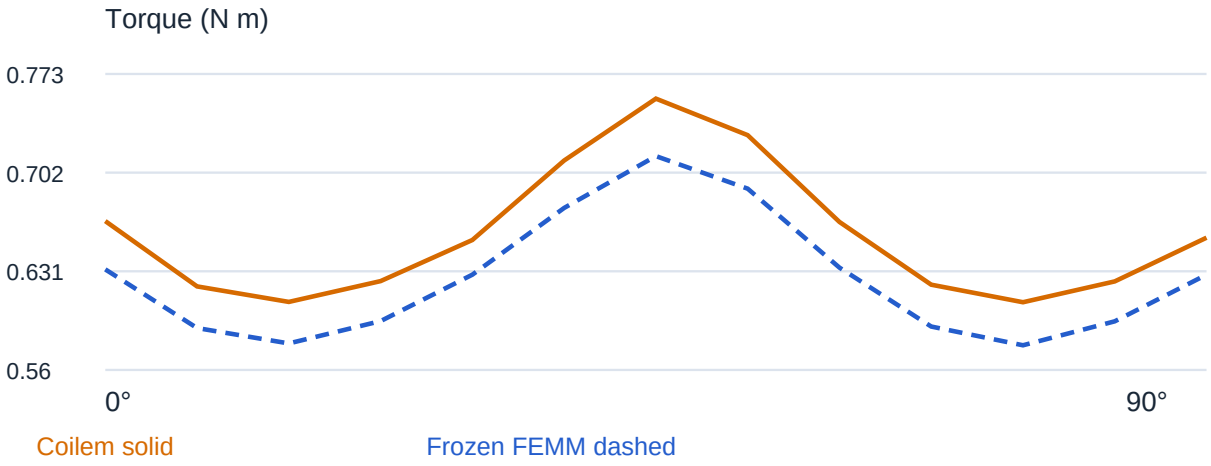


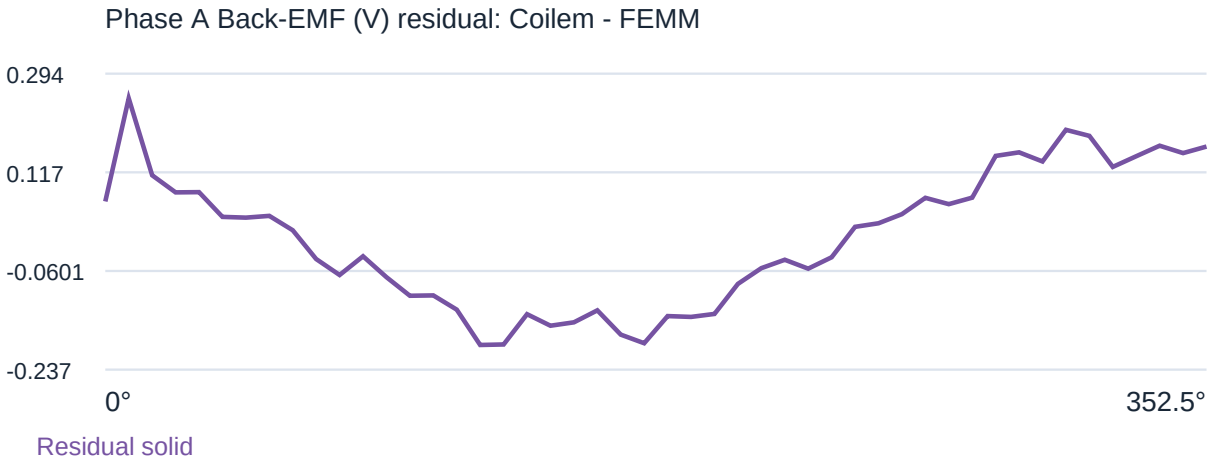

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

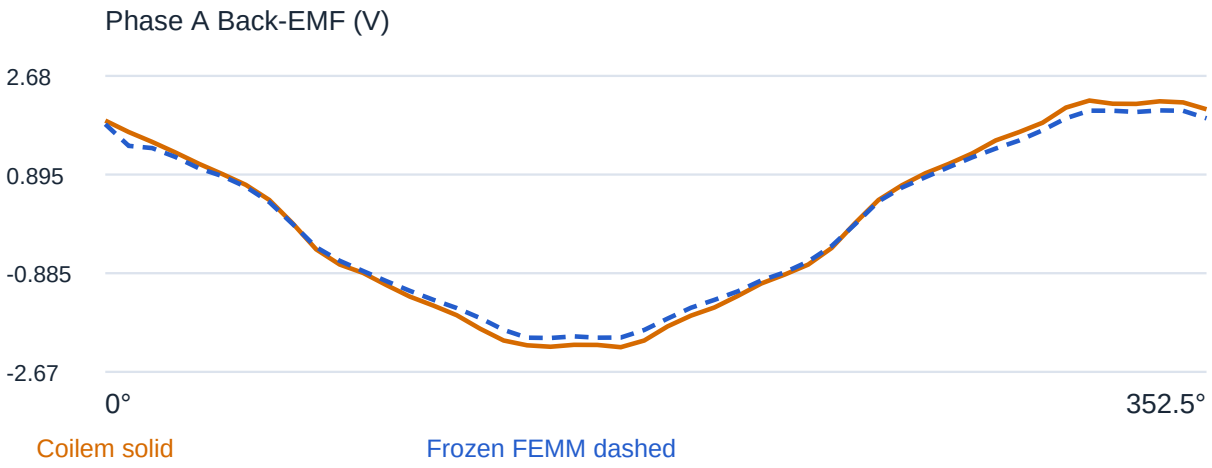
# Phase B – ipm_vshape_8p12s_deep_apex_high_output_v1

Frozen native fixture JSON



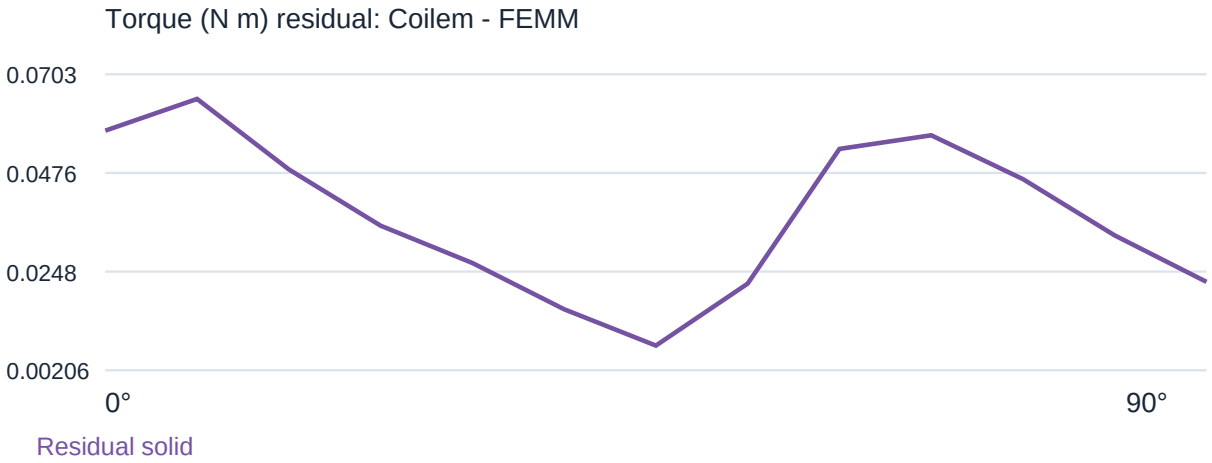



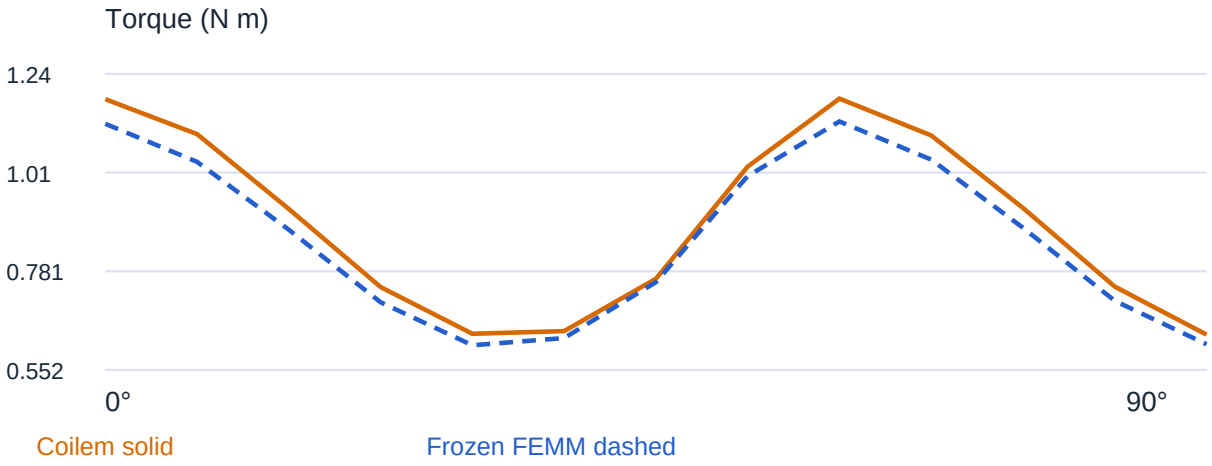


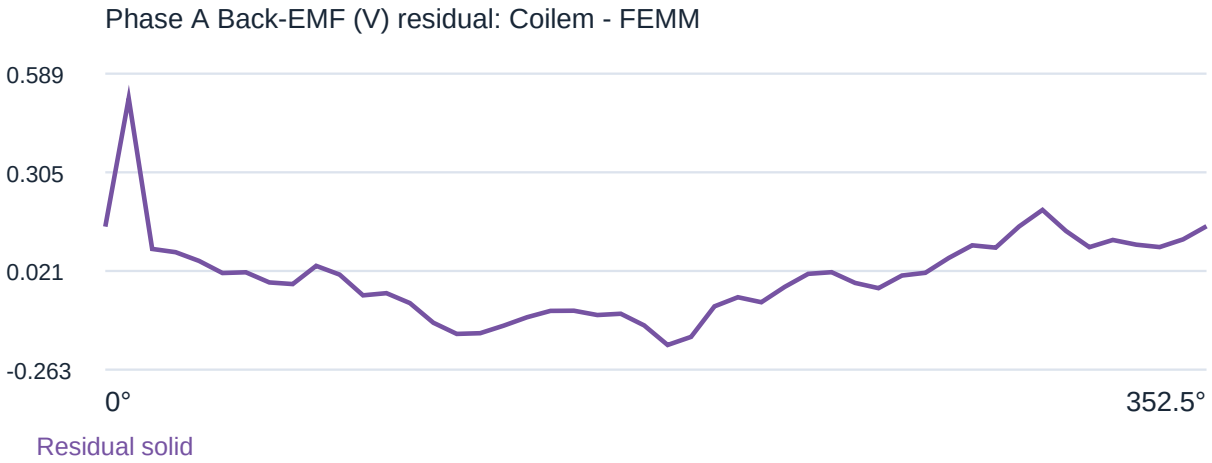

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

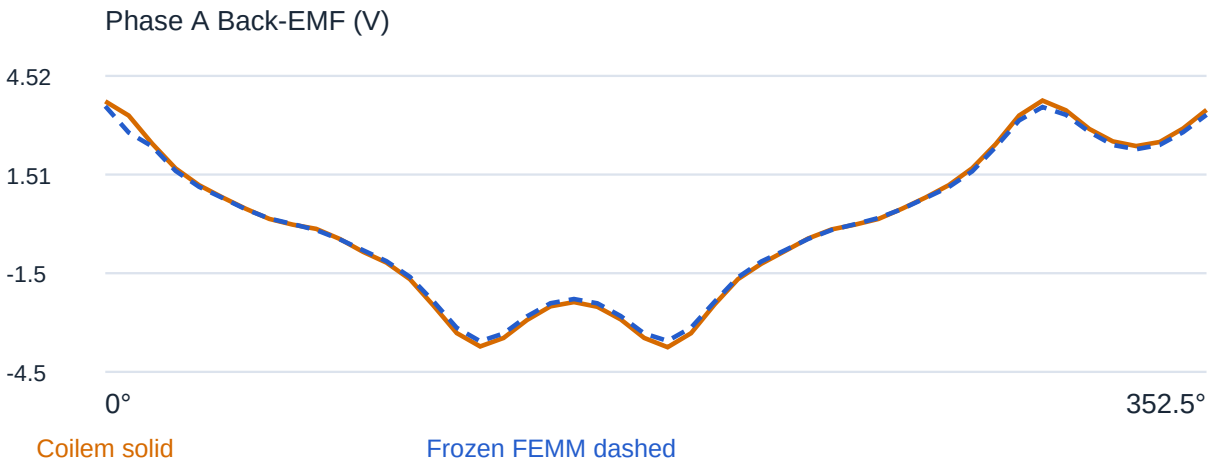
## Phase B – ipm_vshape_8p12s_narrow_angle_contract

Frozen native fixture JSON



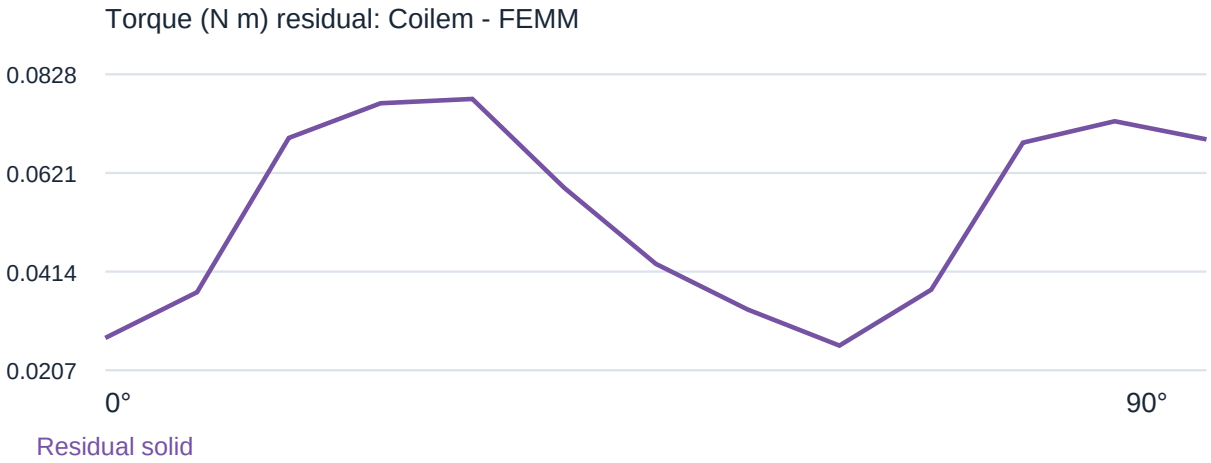





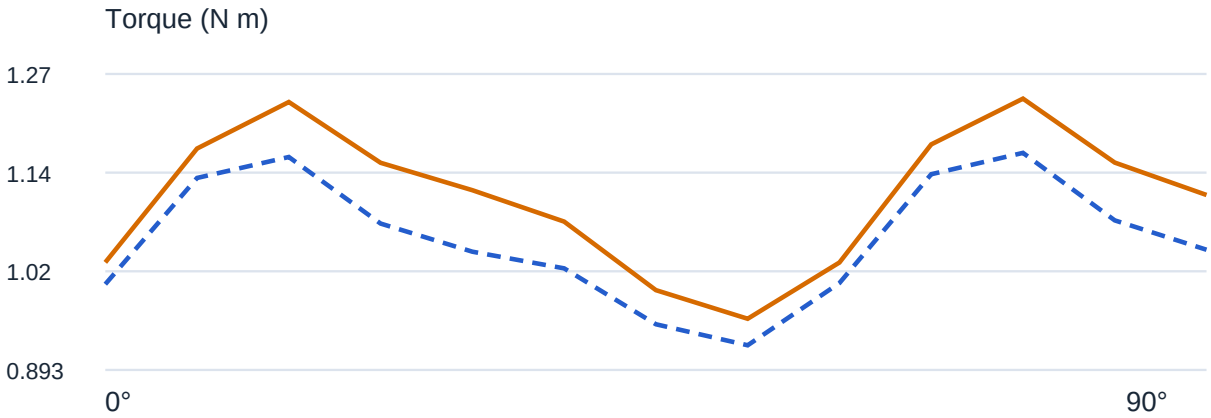

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase B – ipm_vshape_8p12s_pocket_offset_contract

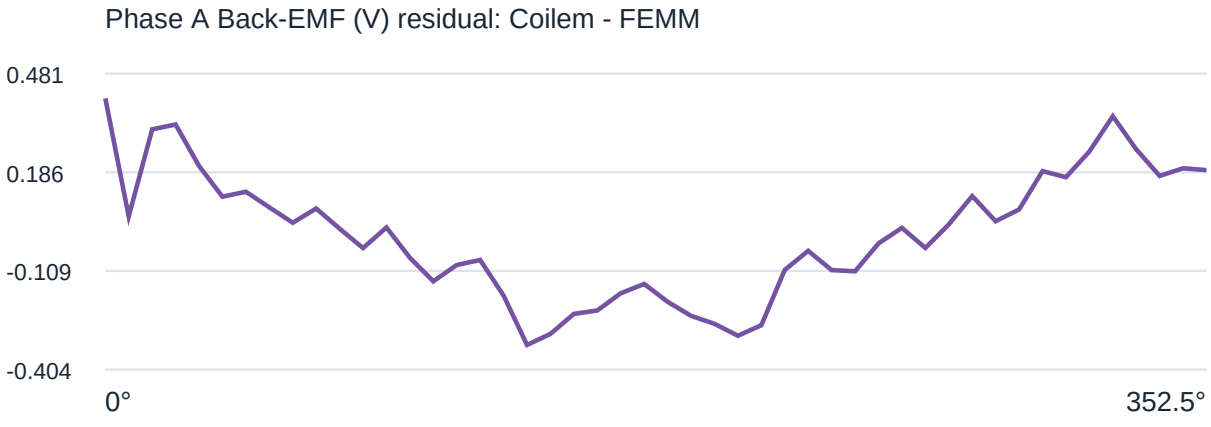
Frozen native fixture JSON



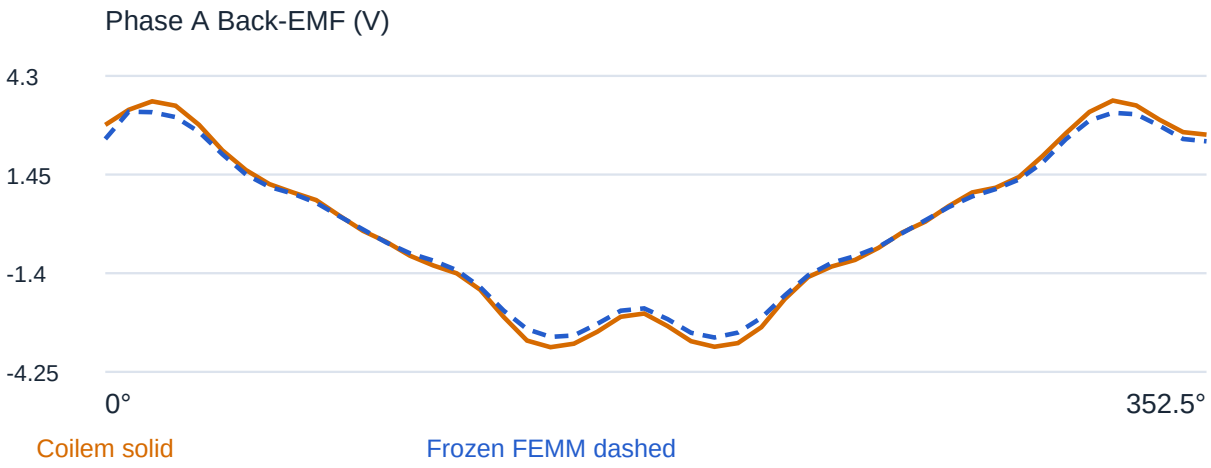

Residual solid




Coilem solid  Frozen FEMM dashed



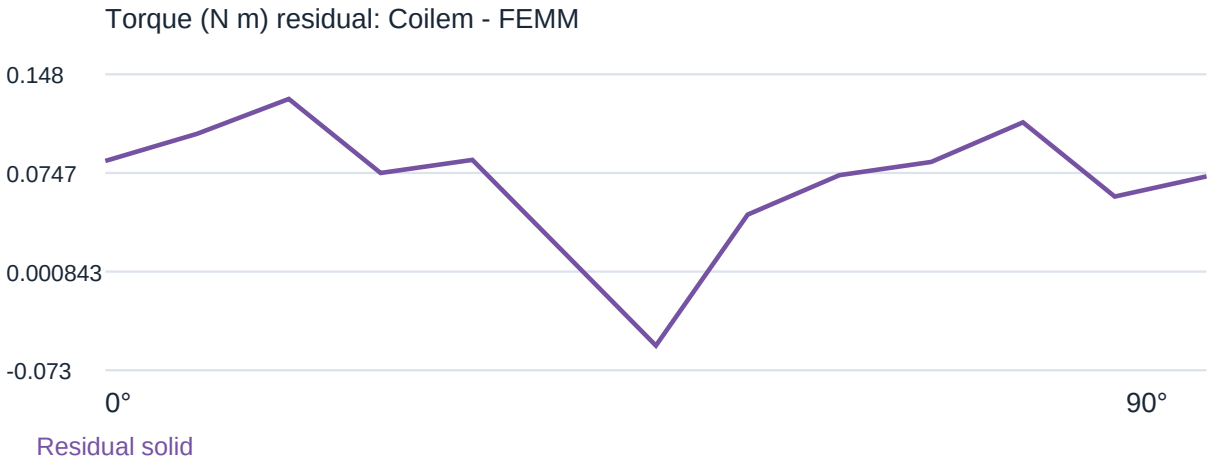

Residual solid

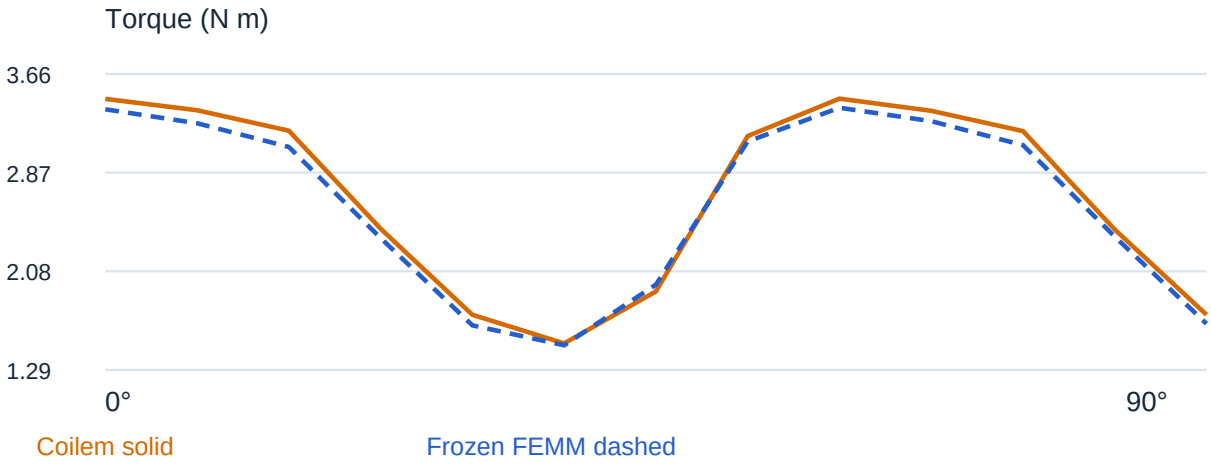


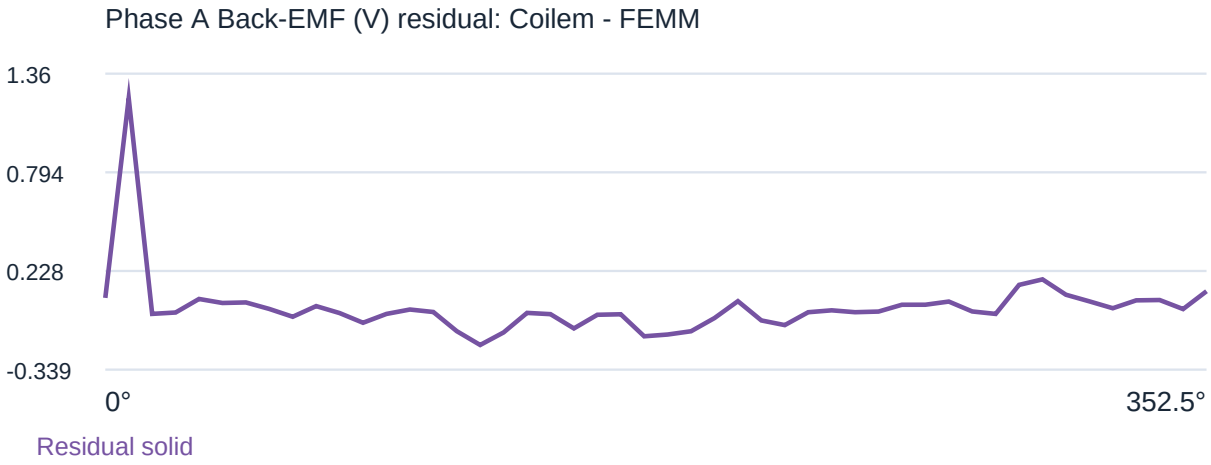

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

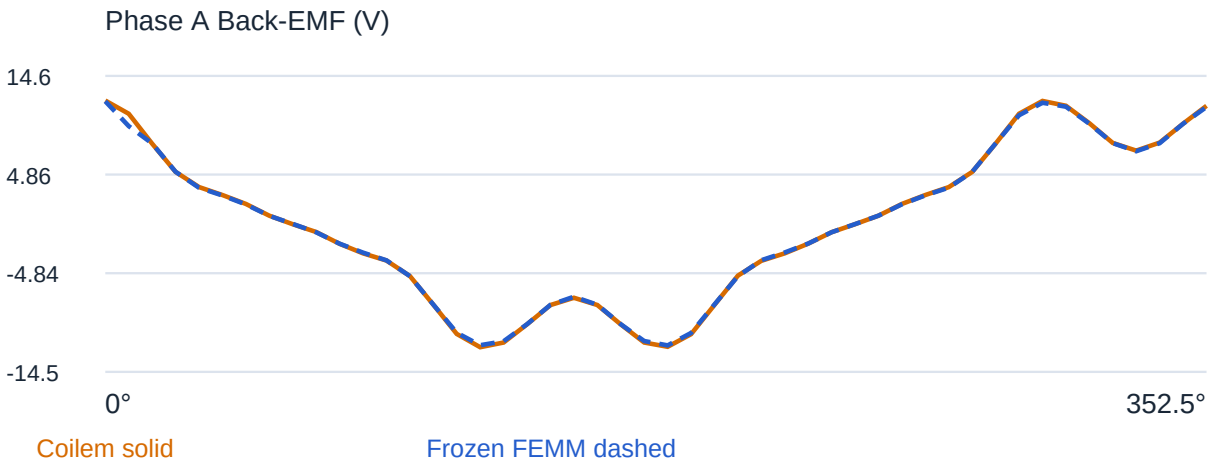
## Phase B – ipm_vshape_8p12s_thin_bridge_contract

Frozen native fixture JSON



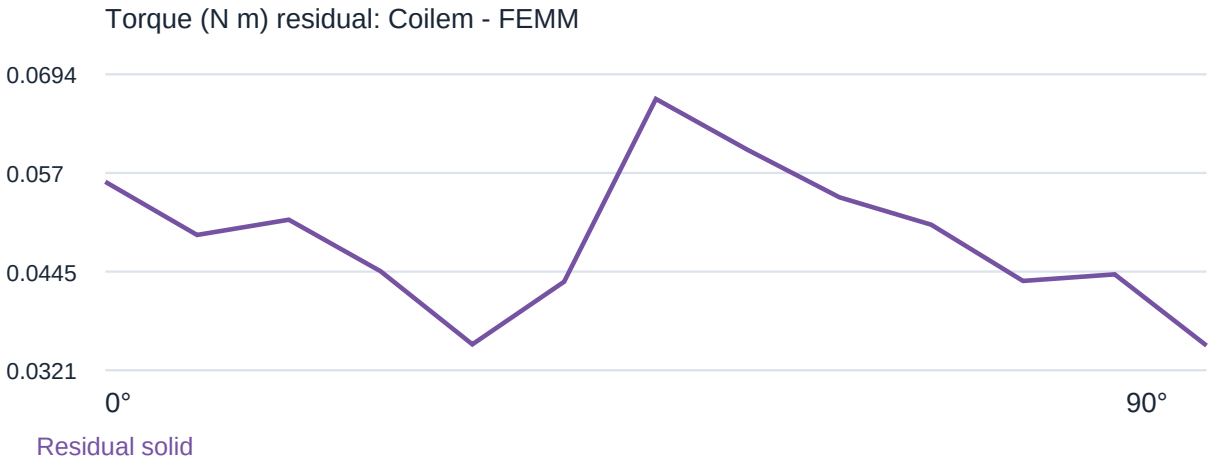





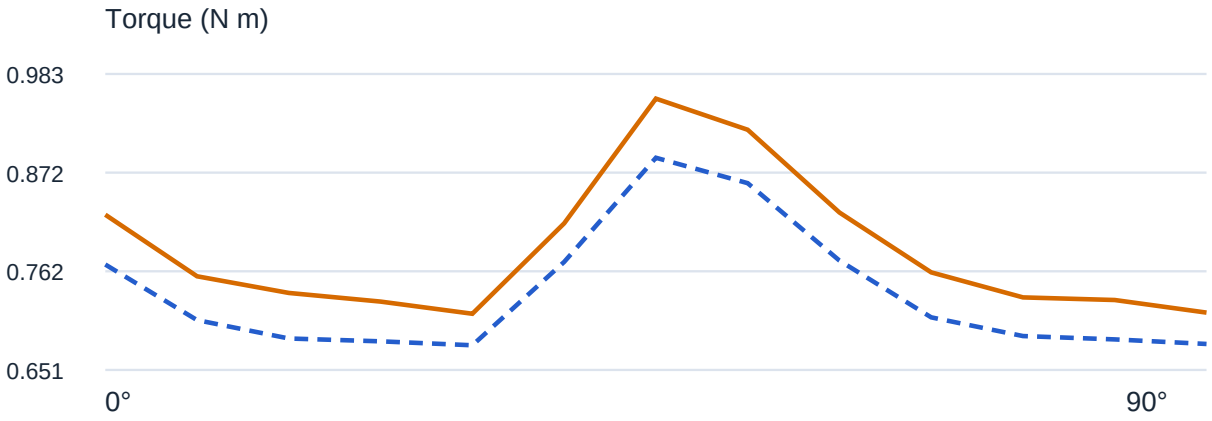

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase B – ipm_vshape_8p12s_wide_angle_high_output_v1

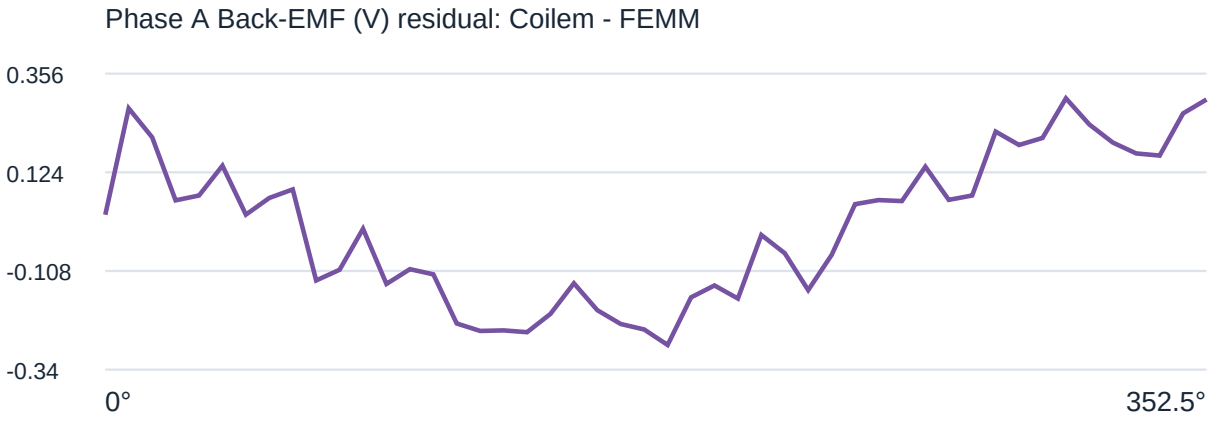
Frozen native fixture JSON



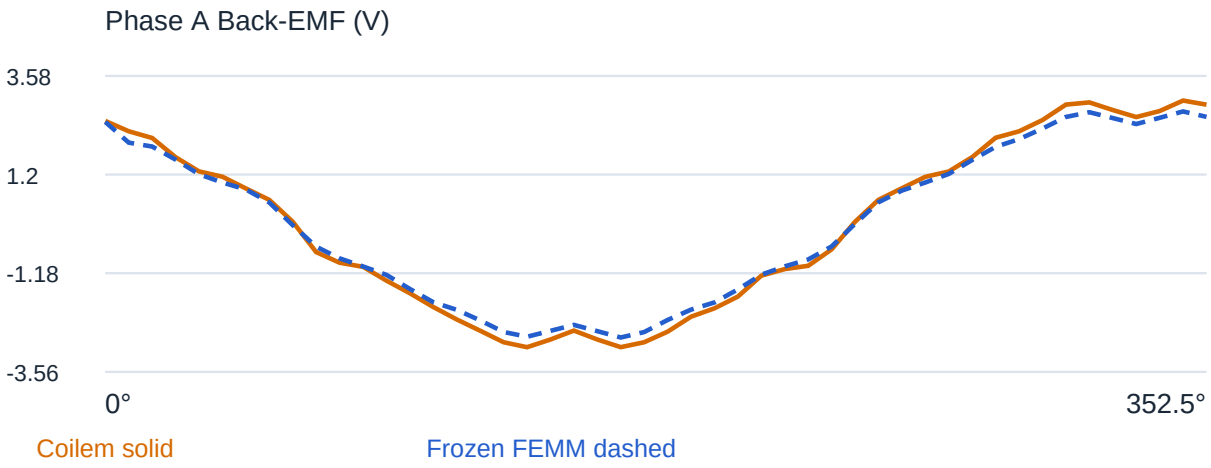

Residual solid




Coilem solid  Frozen FEMM dashed




Residual solid




| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase C – ornl_prius_2004_ipm

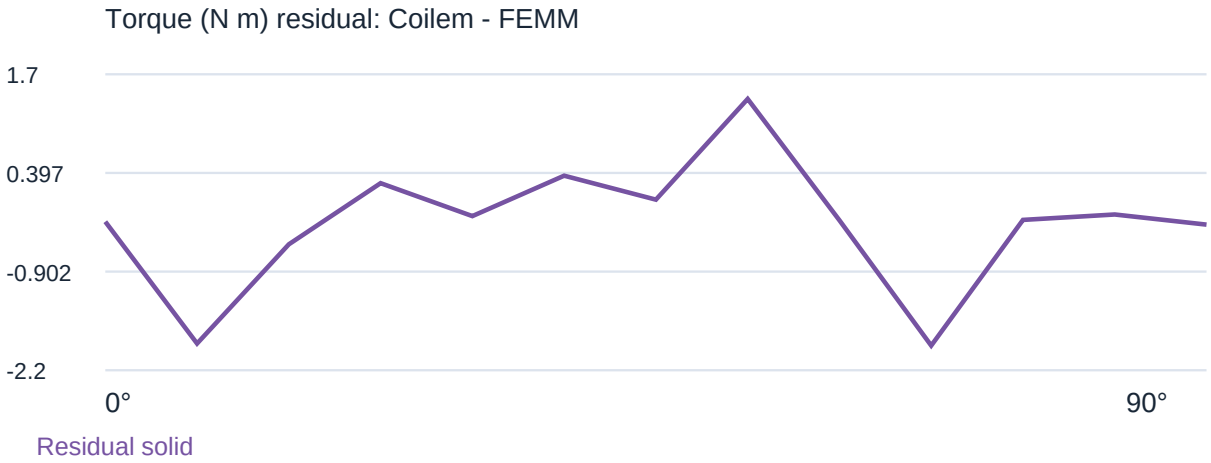
Frozen native fixture JSON

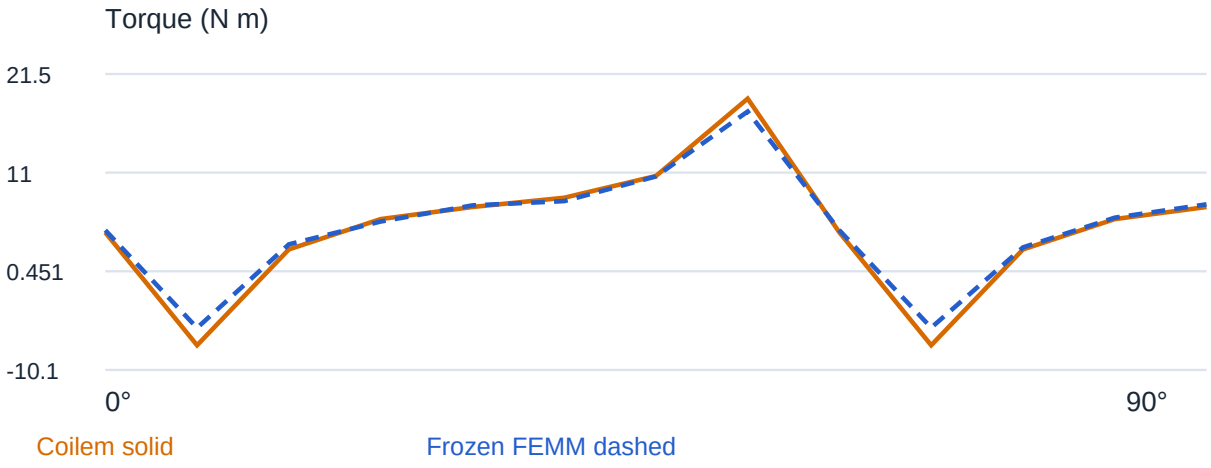
Deferred hardware coverage (v0.5.0). This legacy ORNL-derived HoleM50 fixture remains unsupported in the registered suite. The separate fixed Pyleecan software references have different case IDs and inputs. Resolve the ORNL reconstruction and matched-reference scope separately; Pyleecan software reproduction does not qualify this legacy hardware fixture. Numerical status: UNSUPPORTED; no comparison was performed.

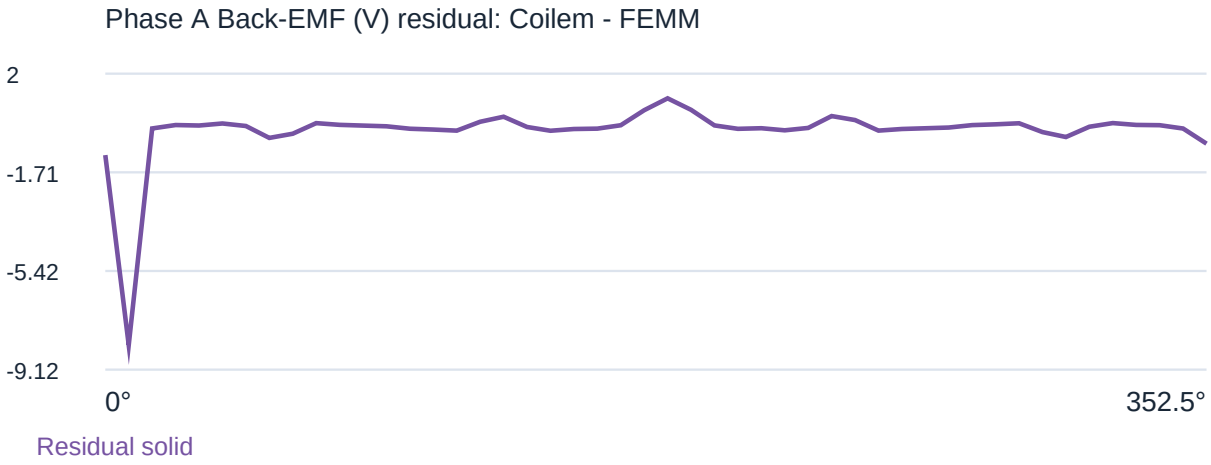
The public launch supports flat-buried and V-shape IPM only.

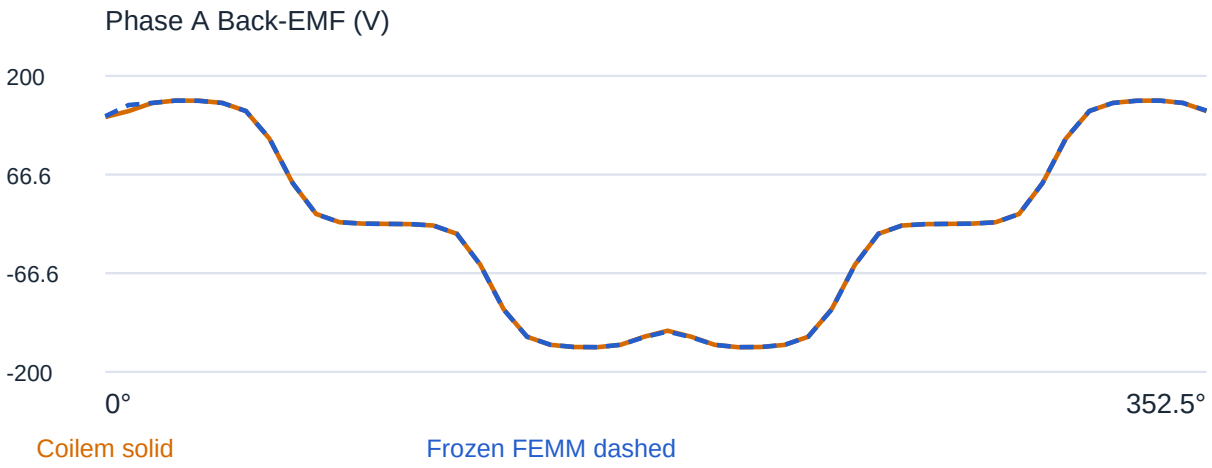
# Phase C – spm_12p36s_distributed_femm_validation

Frozen native fixture JSON



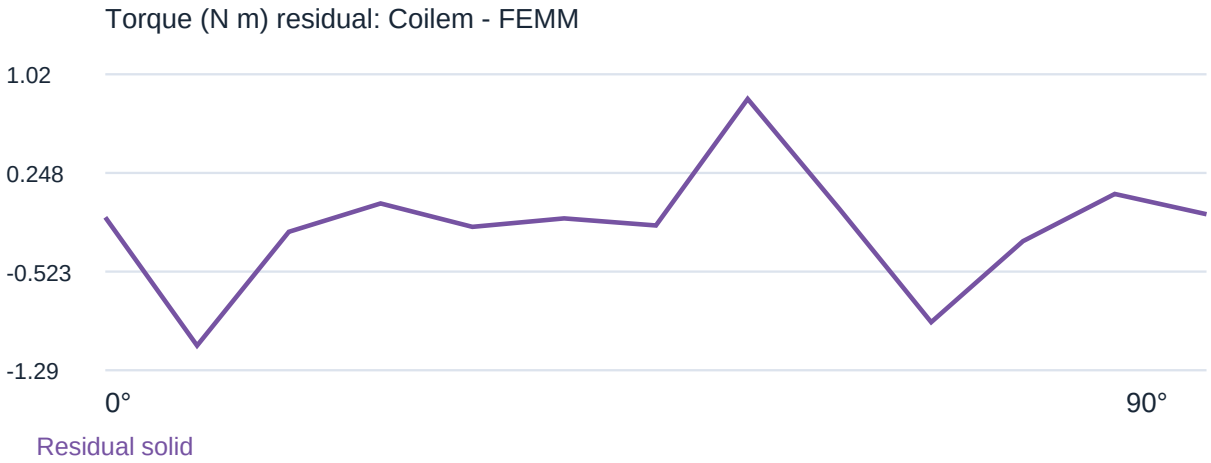



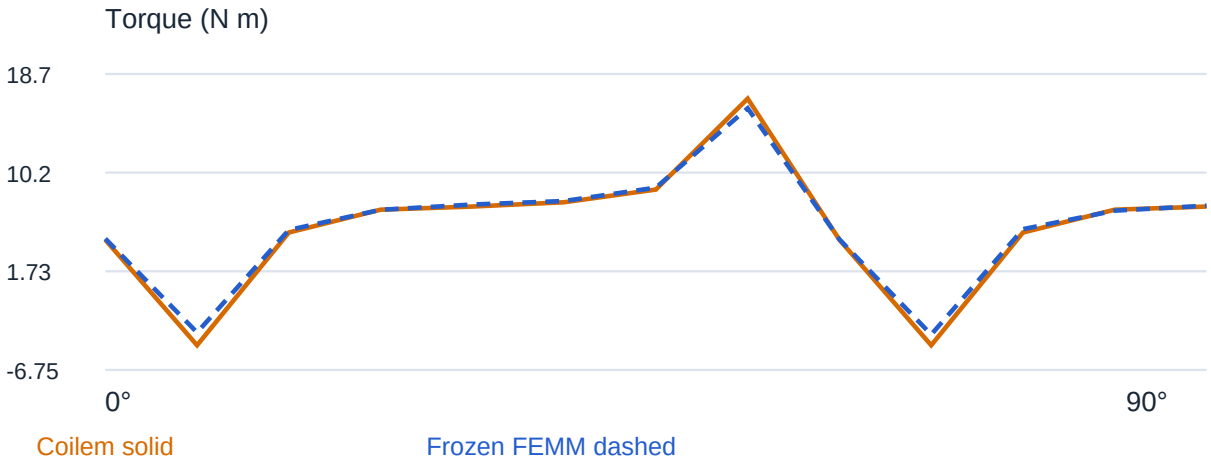


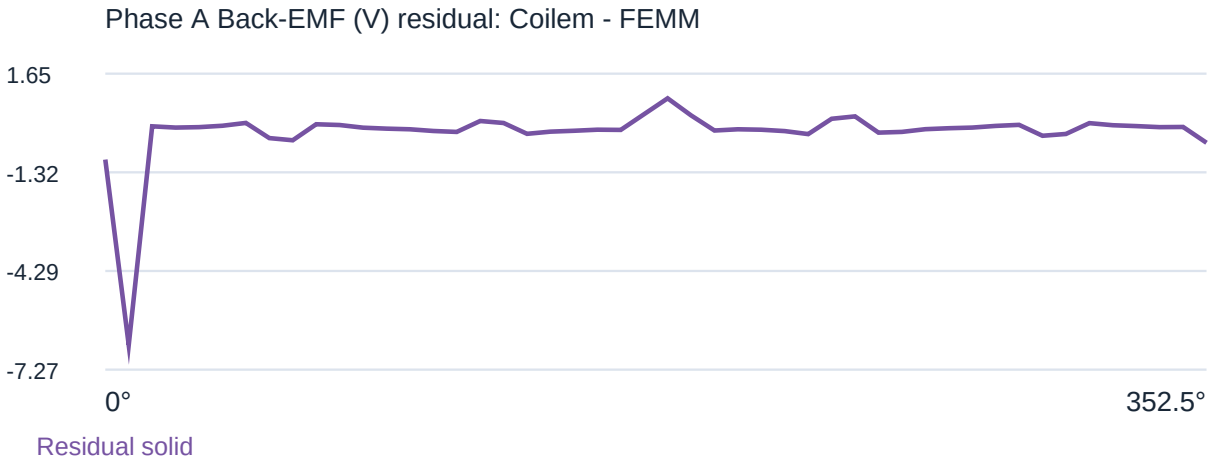

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

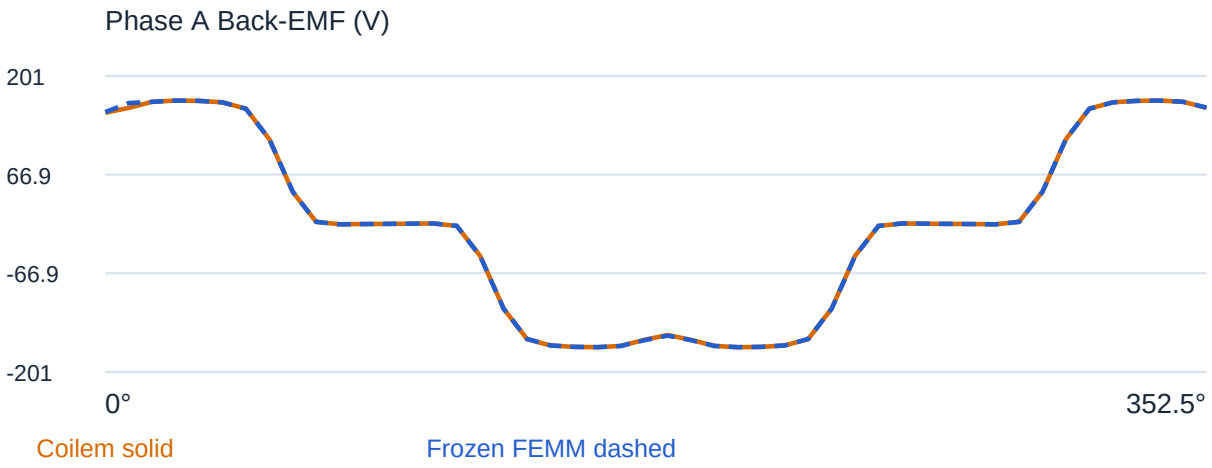
# Phase C – spm_8p24s_distributed_femm_validation

Frozen native fixture JSON



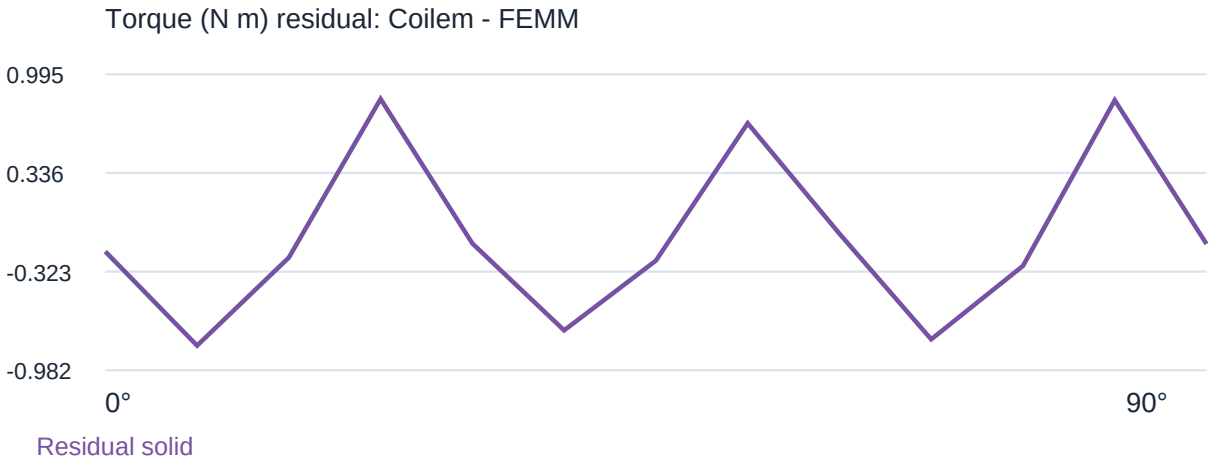





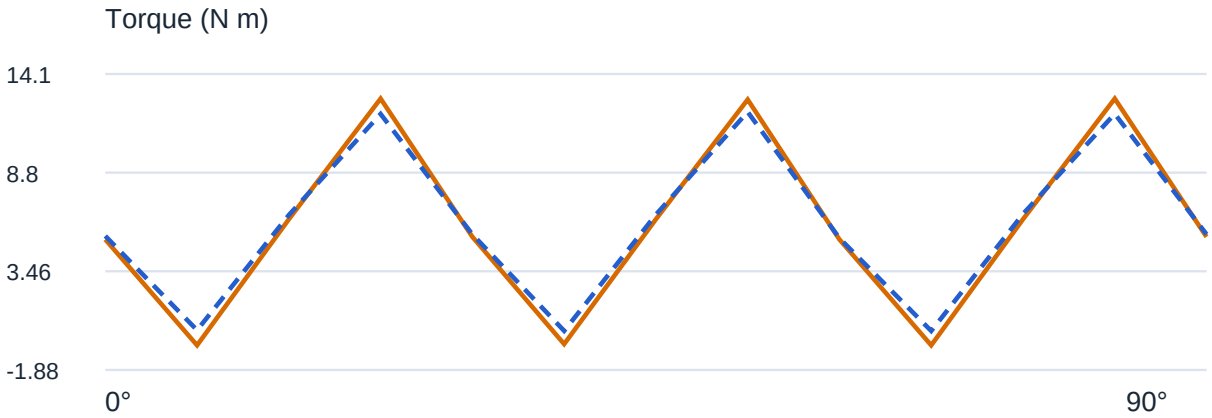

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase C – spm_8p48s_chorded_5_6

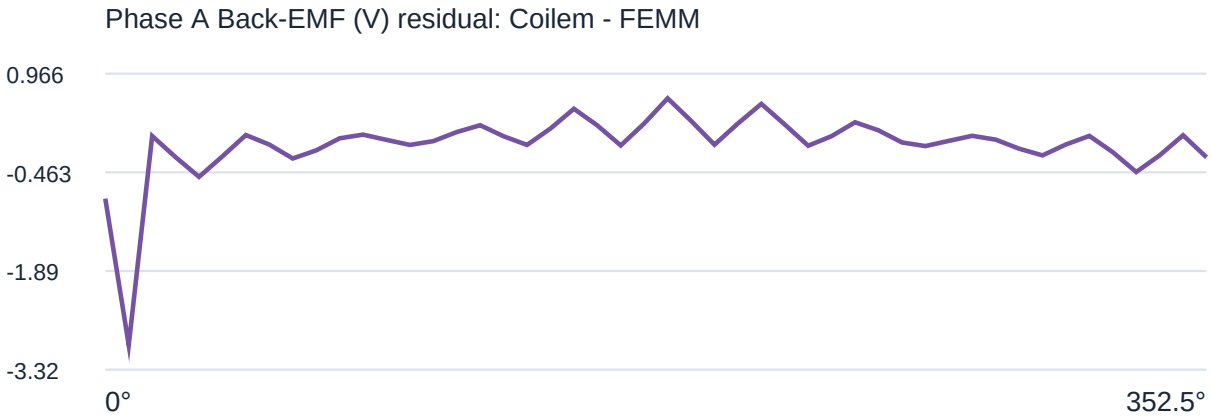
Frozen native fixture JSON



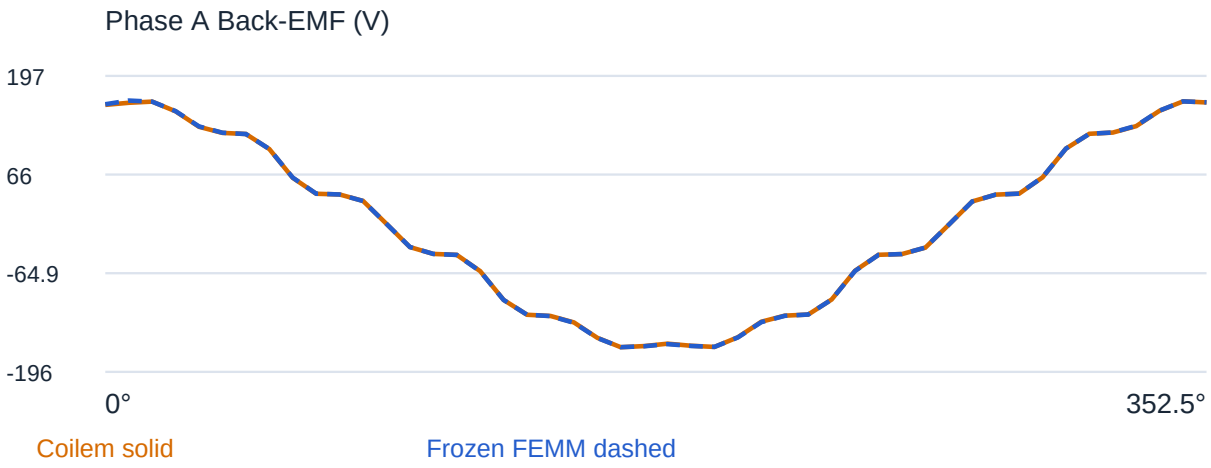

Residual solid




Coilem solid  Frozen FEMM dashed



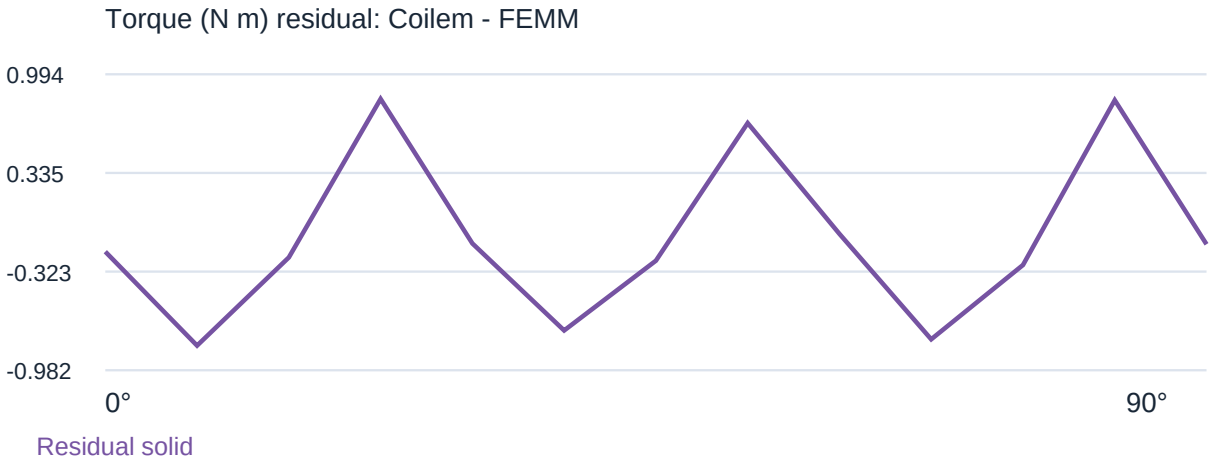

Residual solid



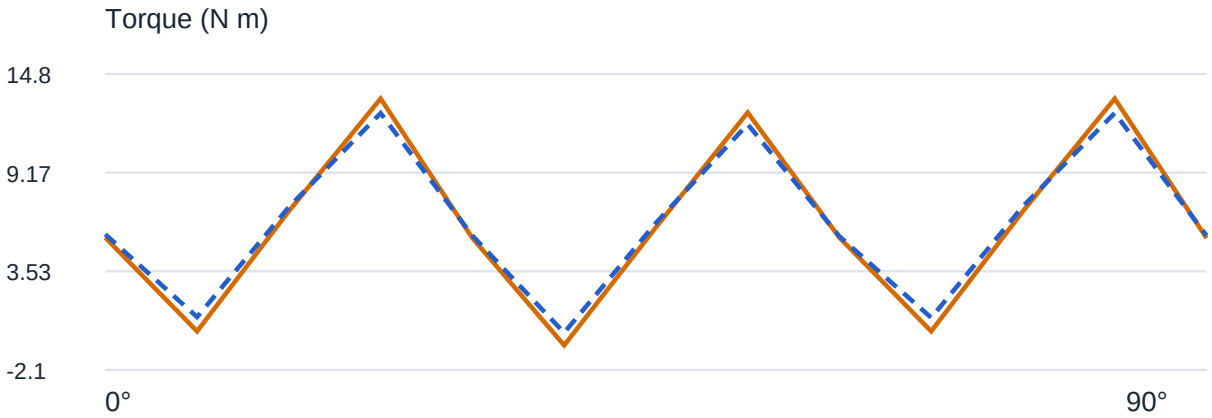

| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

# Phase C – spm_8p48s_distributed_femm_validation

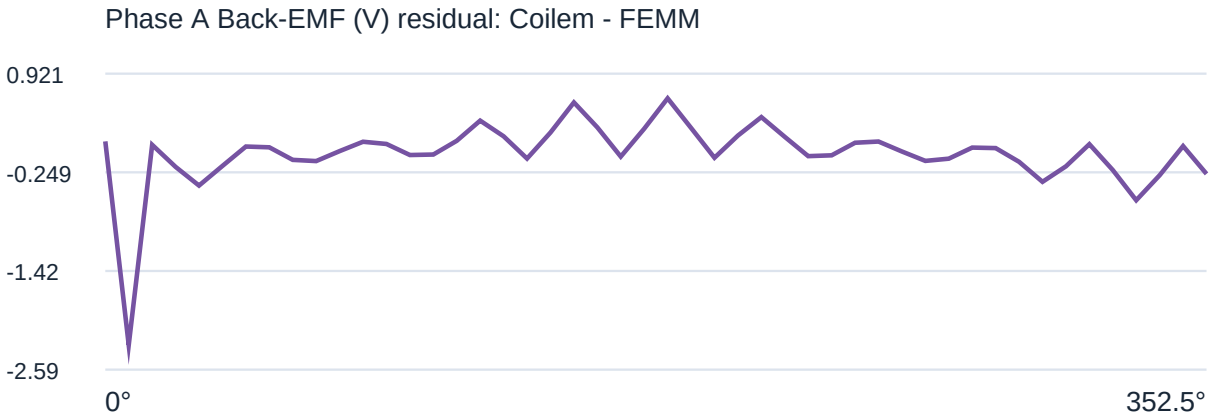
Frozen native fixture JSON



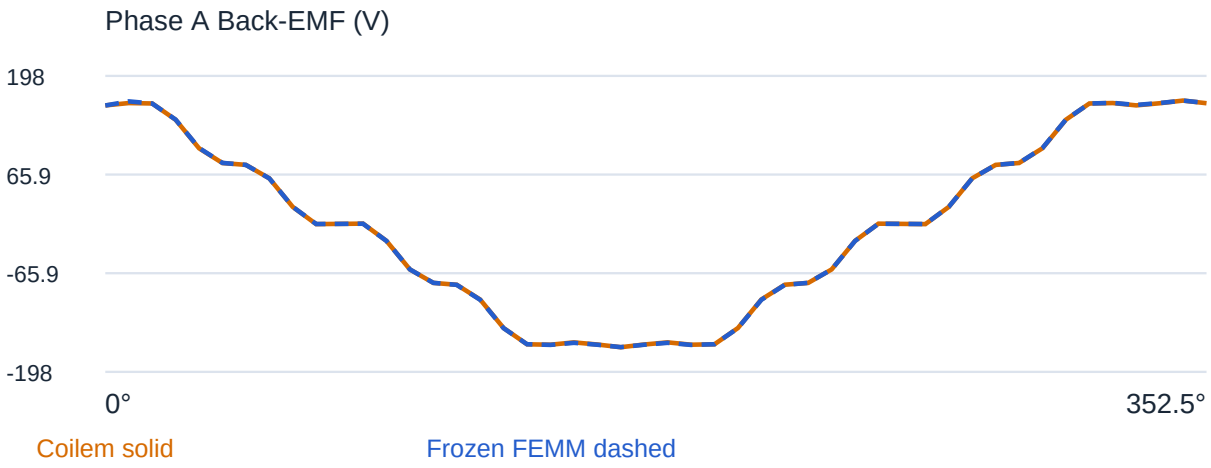

Residual solid




Coilem solid  Frozen FEMM dashed




Residual solid




| Gate | Outcome |
|---|---|
| mean_torque | PASS |
| bemf_fundamental | PASS |
| bemf_peak | PASS |
| torque_method | PASS |
| native_mesh | PASS |
| angle_grids | PASS |

## Phase D – BLDC results

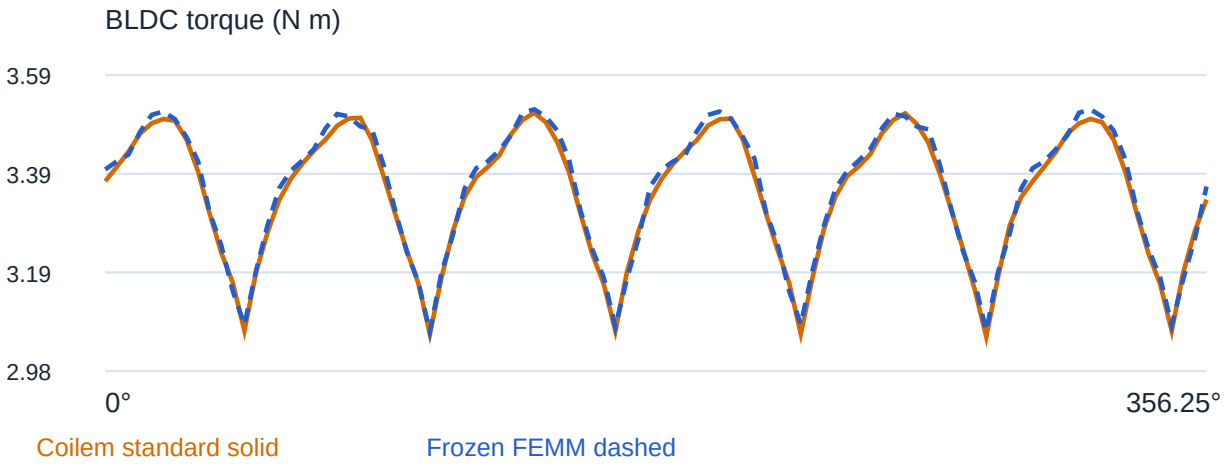
| Fixture | Outcome |
|---|---|
| spm_14p12s_high_pole | PASS |
| spm_4p12s_small | PASS |
| spm_8p12s_reference | PASS |

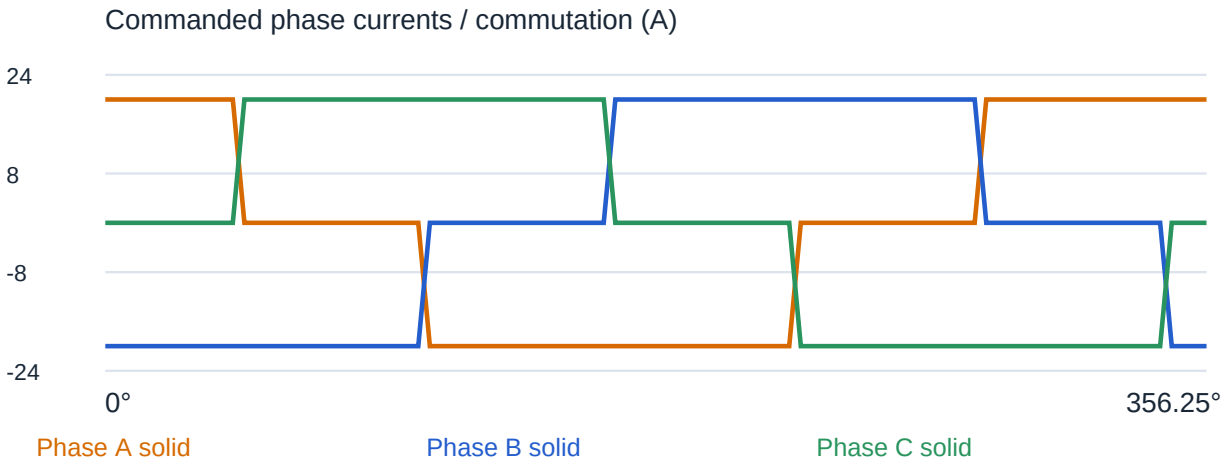
## Phase D – spm_14p12s_high_pole

Frozen native-standard fixture JSON

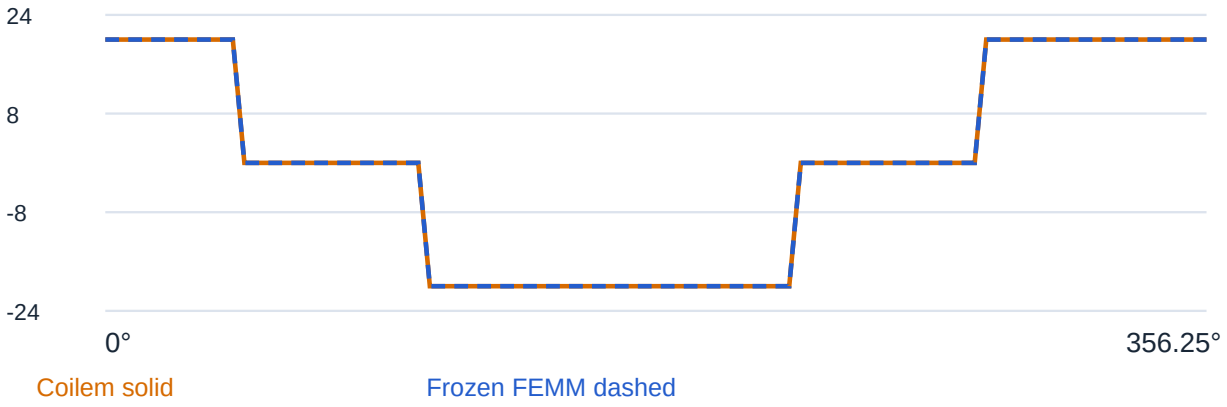
Frozen native-fine fixture JSON

Frozen analytical fixture JSON



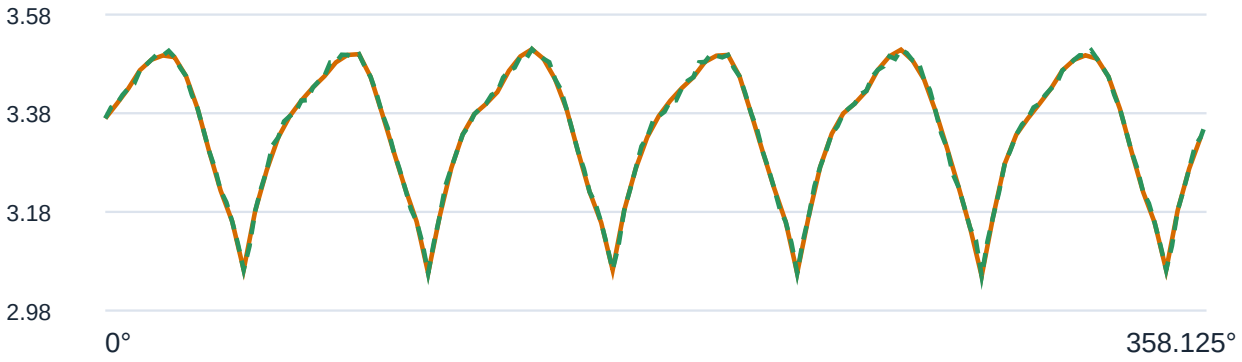



Phase A current: native / frozen reference (A)



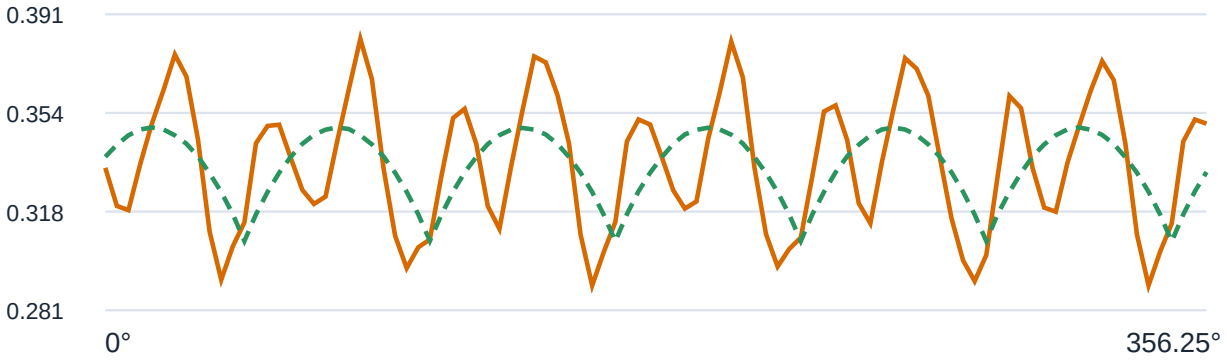

Coilem solid  Frozen FEMM dashed

Angular convergence: normal spatial mesh (N m)




Standard solid  Fine angular dashed

Low-current power / speed and field torque (N m); gate uses cycle means




Field torque solid  Power / speed dashed

| Gate | Outcome |
|---|---|
| mean_torque_delta_pct | PASS |
| torque_waveform_nrmse_pct | PASS |
| commutation_transition_delta_samples | PASS |
| power_torque_delta_pct | PASS |
| native_standard_to_fine_mean_torque_delta_pct | PASS |
| native_standard_to_fine_ripple_delta_percentage_points | PASS |

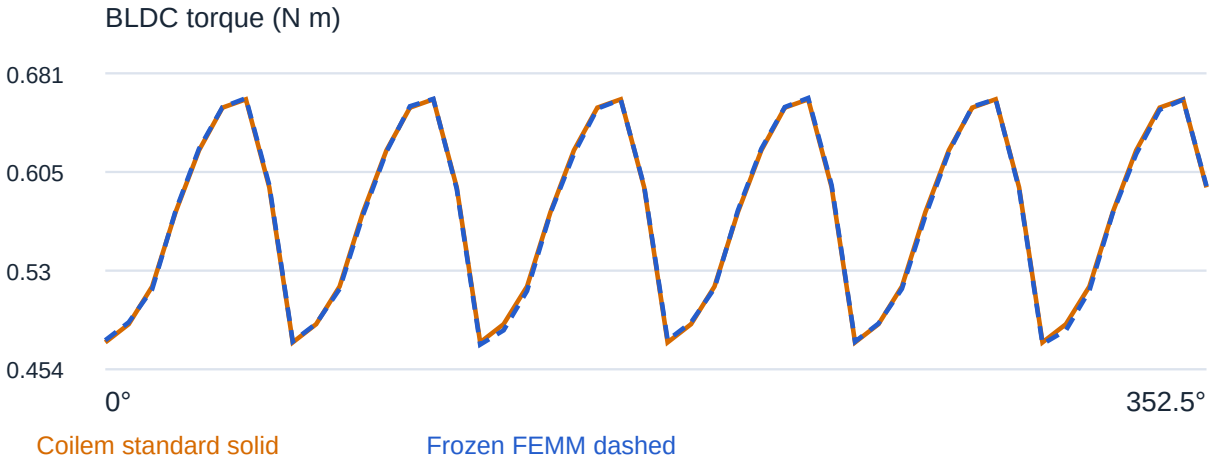
| Gate | Outcome |
|---|---|
| native_standard_current_properties | PASS |
| native_fine_current_properties | PASS |
| femm_standard_current_properties | PASS |
| analytical_low_current_current_properties | PASS |
| native-standard_golden_current_sequence | PASS |
| native-fine_golden_current_sequence | PASS |
| analytical_golden_current_sequence | PASS |
| femm-standard_golden_current_sequence | PASS |

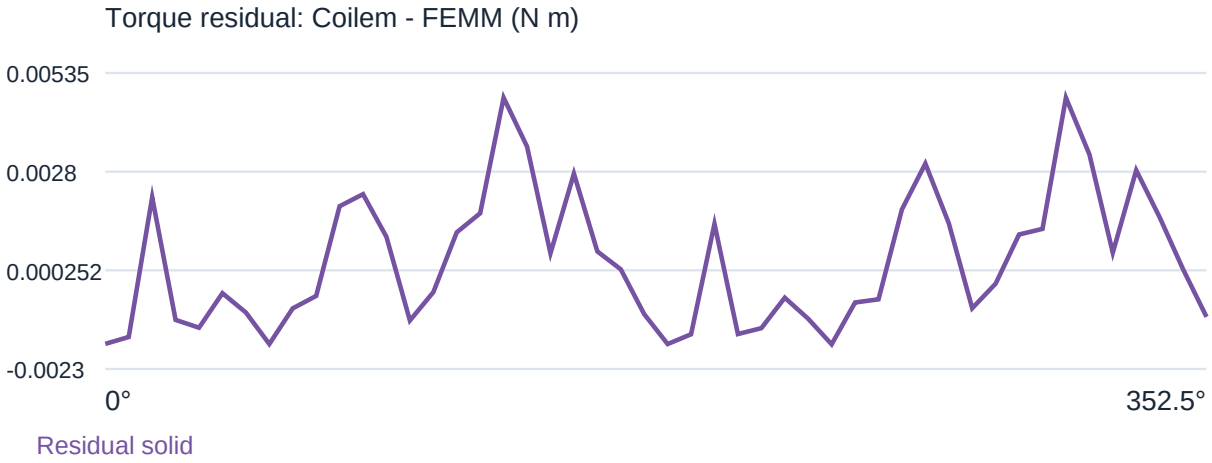
# Phase D – spm_4p12s_small

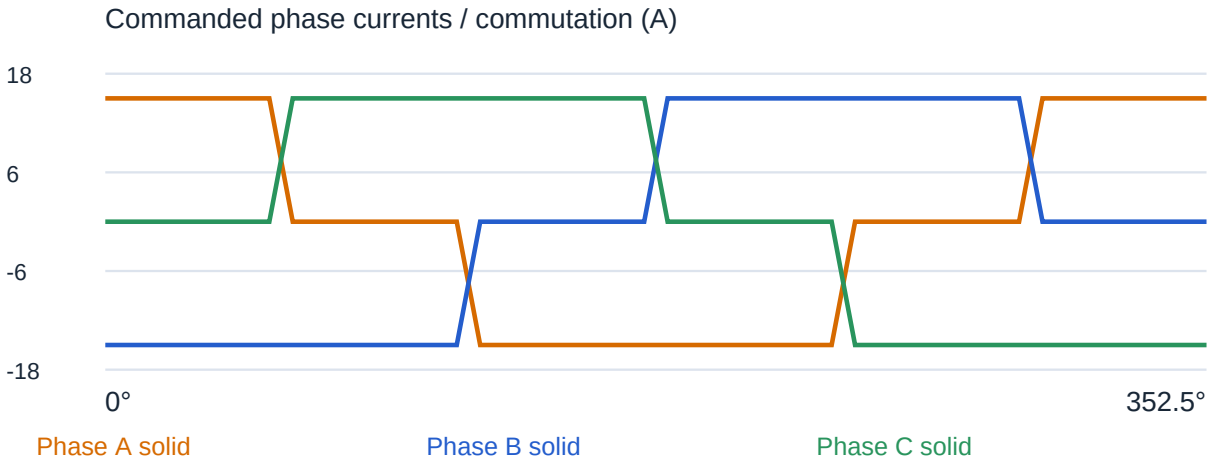
Frozen native-standard fixture JSON

Frozen native-fine fixture JSON

Frozen analytical fixture JSON



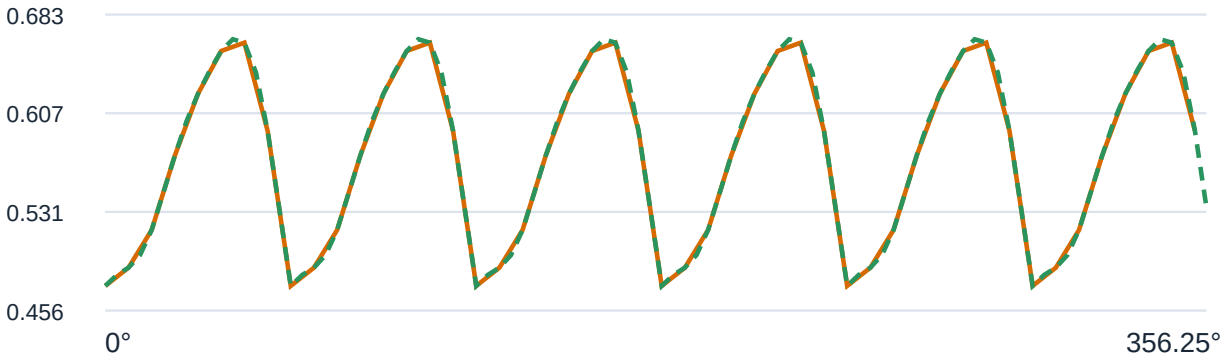



Phase A current: native / frozen reference (A)



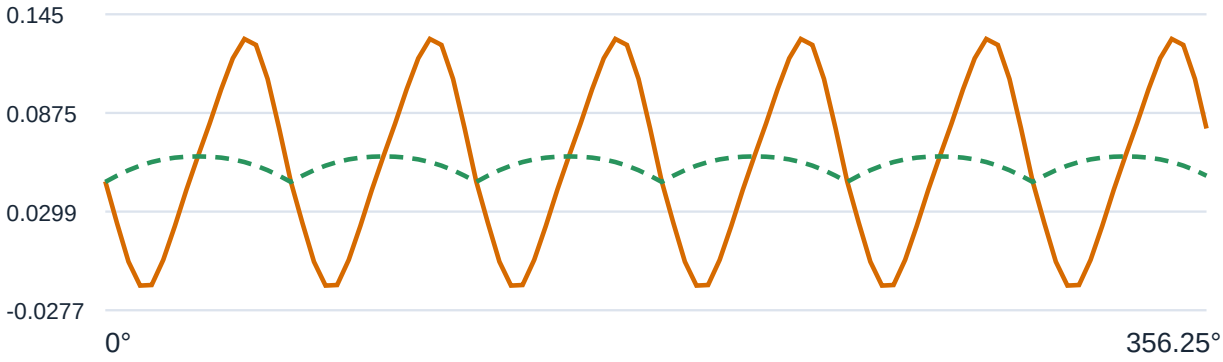

Coilem solid  Frozen FEMM dashed

Angular convergence: normal spatial mesh (N m)




Standard solid  Fine angular dashed

Low-current power / speed and field torque (N m); gate uses cycle means




Field torque solid  Power / speed dashed

| Gate | Outcome |
|---|---|
| mean_torque_delta_pct | PASS |
| torque_waveform_nrmse_pct | PASS |
| commutation_transition_delta_samples | PASS |
| power_torque_delta_pct | PASS |
| native_standard_to_fine_mean_torque_delta_pct | PASS |
| native_standard_to_fine_ripple_delta_percentage_points | PASS |

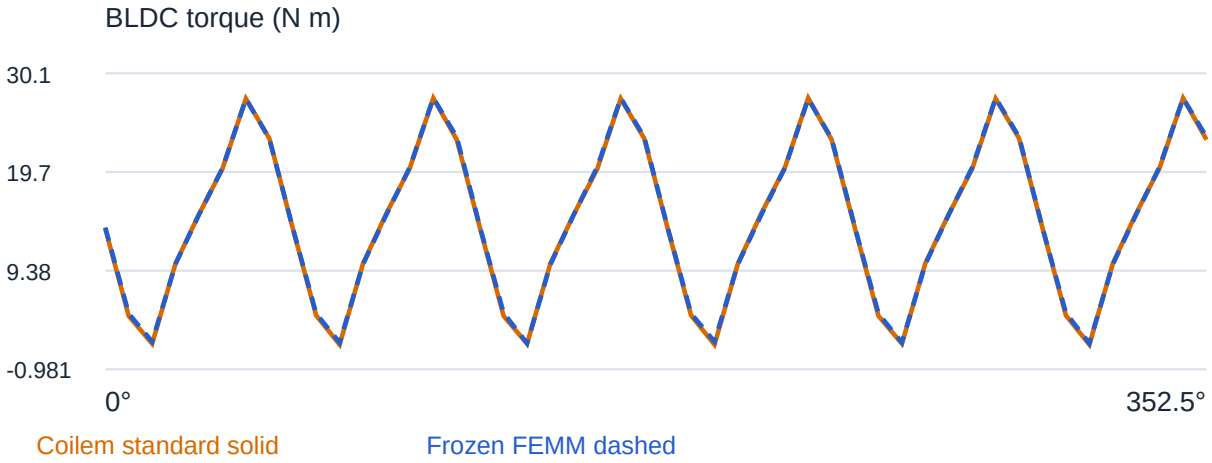
| Gate | Outcome |
|---|---|
| native_standard_current_properties | PASS |
| native_fine_current_properties | PASS |
| femm_standard_current_properties | PASS |
| analytical_low_current_current_properties | PASS |
| native-standard_golden_current_sequence | PASS |
| native-fine_golden_current_sequence | PASS |
| analytical_golden_current_sequence | PASS |
| femm-standard_golden_current_sequence | PASS |

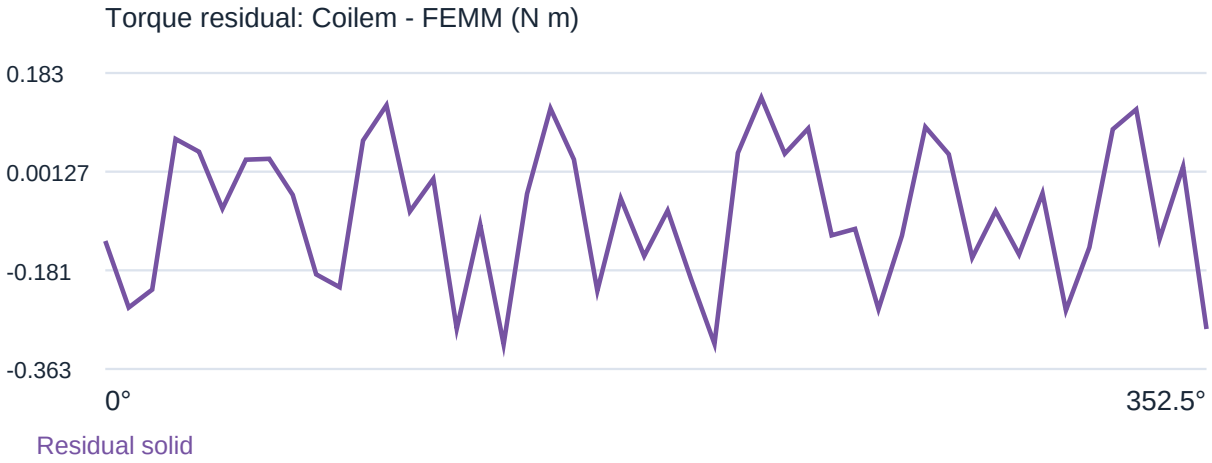
# Phase D – spm_8p12s_reference

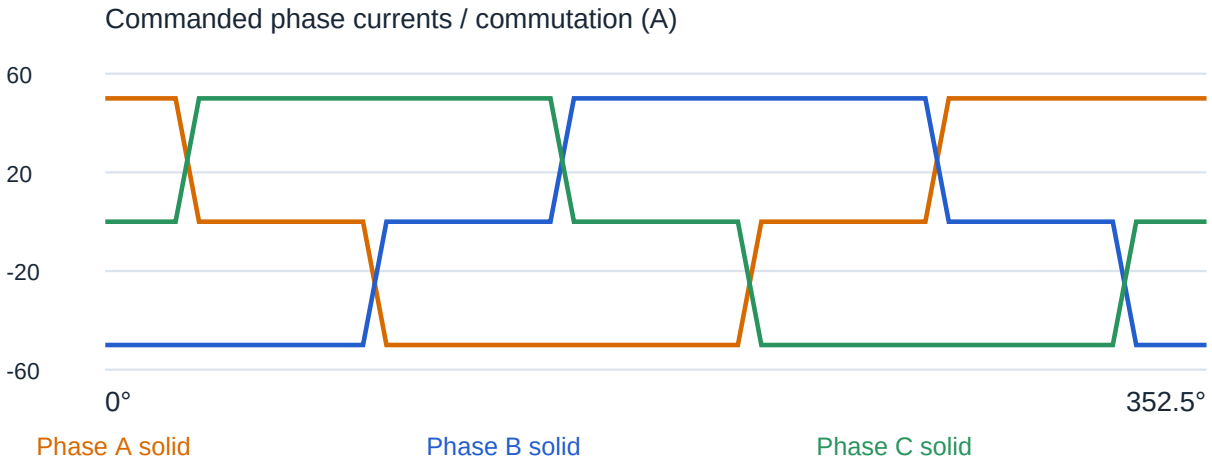
Frozen native-standard fixture JSON

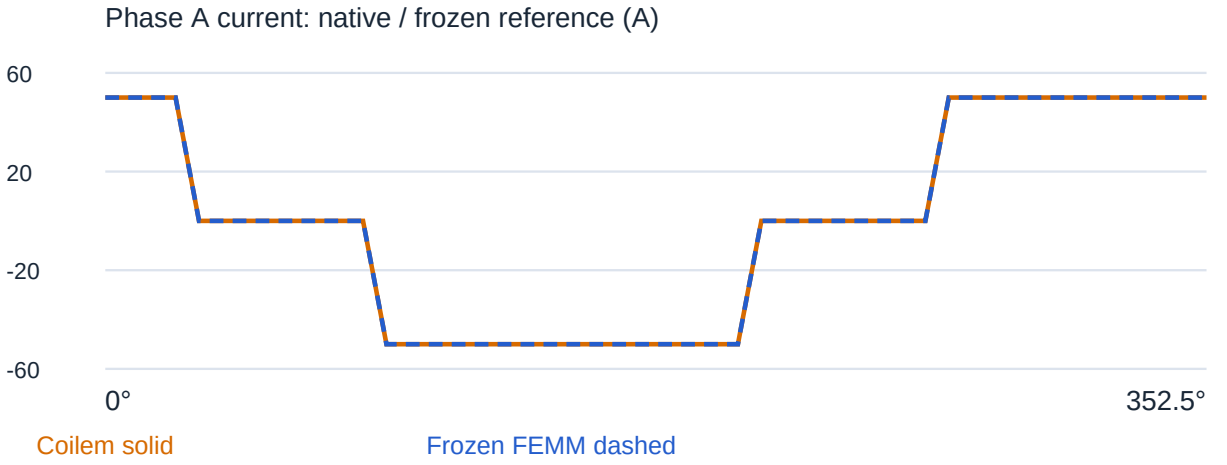
Frozen native-fine fixture JSON

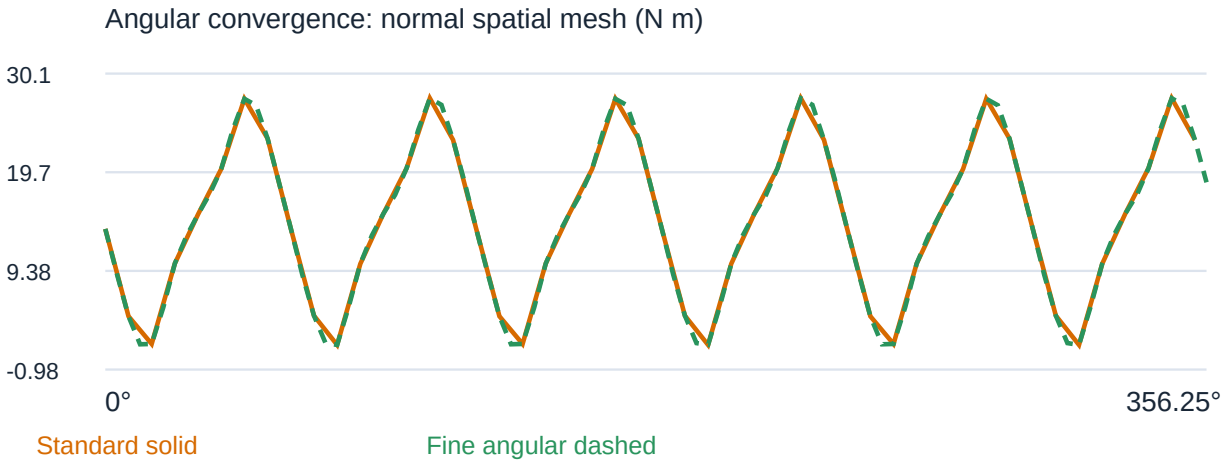
Frozen analytical fixture JSON

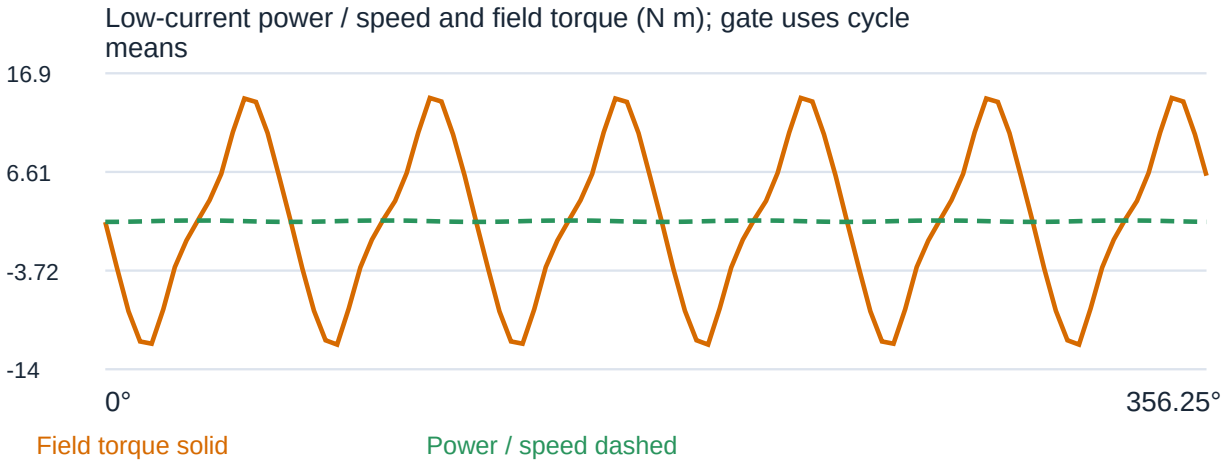





| Gate | Outcome |
|---|---|
| mean_torque_delta_pct | PASS |
| torque_waveform_nrmse_pct | PASS |
| commutation_transition_delta_samples | PASS |
| power_torque_delta_pct | PASS |
| native_standard_to_fine_mean_torque_delta_pct | PASS |
| native_standard_to_fine_ripple_delta_percentage_points | PASS |

| Gate | Outcome |
|---|---|
| native_standard_current_properties | PASS |
| native_fine_current_properties | PASS |
| femm_standard_current_properties | PASS |
| analytical_low_current_current_properties | PASS |
| native-standard_golden_current_sequence | PASS |
| native-fine_golden_current_sequence | PASS |
| analytical_golden_current_sequence | PASS |
| femm-standard_golden_current_sequence | PASS |